

WELCOME TO SUGARCUBE™



Thank You for purchasing the SugarCube **SC2-Plus**.

Your experience of high-end audio is about to get a lot sweeter when all your vinyl will sound like mint.

These pages provide an overview of the set-up and control of the **SC2-Plus**.

For any usage questions,
please contact:
support@sweetvinyl.com



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Section 1

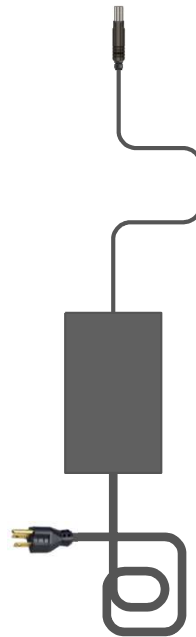
Unpacking Your SugarCube



1.1 Unpacking Your SugarCube



SugarCube SC2-Plus Unit



**AC Power Adapter,
5V, 4A DC Output**



Wifi USB Adapter



**USB
Stick**

Please download the **SugarCube App** from your mobile device's app store or use a **Browser** to control your **SugarCube** remotely.

This adapter connects the SugarCube to your WiFi network or you can use a wired Internet connection if available.

USB-stick contains the full SugarCube Setup and User Guide.

1.2 SC2-Plus Front-Panel Controls

VOLUME

Controls the **VOLUME** output of the Headphone when headphone is unmuted, and unit is not in **BYPASS**.

PAIR FUNCTION

If asked for a pairing code (Pair), press to link your device with this SugarCube. Instructions found in the user guide.

RECORD

When pressed, SugarCube is recording audio. The **Ready/Rec** LED will turn from green to red.

STORAGE

USB thumb drive port accepts up to USB 3.0 media for transferring recordings. Instructions found in the manual.



HEADPHONE

When pressed and the LED is lit, The **HEADPHONE** is muted. When unmuted the headphone jack outputs the signal processed by the SugarCube.

BYPASS

When pressed and the LED is lit, an internal relay has shunted the **LINE IN** directly to **LINE OUT** completely bypassing all internal circuitry.

STRENGTH

Turn knob to control the strength of the click and pop detection and removal process. Almost all LPs will sound great at a setting of 4 or 5.

Note: we recommend that you do **NOT** adjust **STRENGTH** all the way to **10**, you will rarely need this much. Sometimes less is more.

In (engage remove)

When pressed and the LED is lit, SugarCube is processing audio to remove clicks and pops. The **BYPASS** LED will go out.

CLICK MON

When pressed and the LED is lit, the SugarCube suppresses the music so that you can better hear the clicks and pops. This helps set the **STRENGTH** level for particularly noisy LPs.

1.3 SC2-Plus Back-Panel

DIGITAL OUTPUT

Toslink and SPDIF outputs at 192K/24bit.

DC POWER IN

Connect the supplied adapter to this port.

POWER ON/OFF

Note: when power is off, an internal relay bypasses **LINE IN** to **LINE OUT**.



LINE OUTPUT

Outputs to your line stage (pre-amp). When the front panel **BYPASS** button is pressed, the audio from **LINE IN** passes directly to this **LINE OUT** through a mechanical audiophile-grade relay.

LINE INPUT

Accepts the output of your phono pre-amp. Please do NOT directly plug your turntable into this port unless it has a built-in phono pre-amp with a line-level output.

USB COMPUTER PORT

Use this port to link your computer or music server to for enhanced ADC and DAC functionality. Detailed instructions later in guide.

WIRED LAN

If an Ethernet LAN cable is available, connect to this port instead of plugging in the WiFi adapter.

USB PORT FOR WIFI ADAPTER

Do either of the above when controlling the SC1-Plus on the SugarCube iOS or Android app or from your desktop.

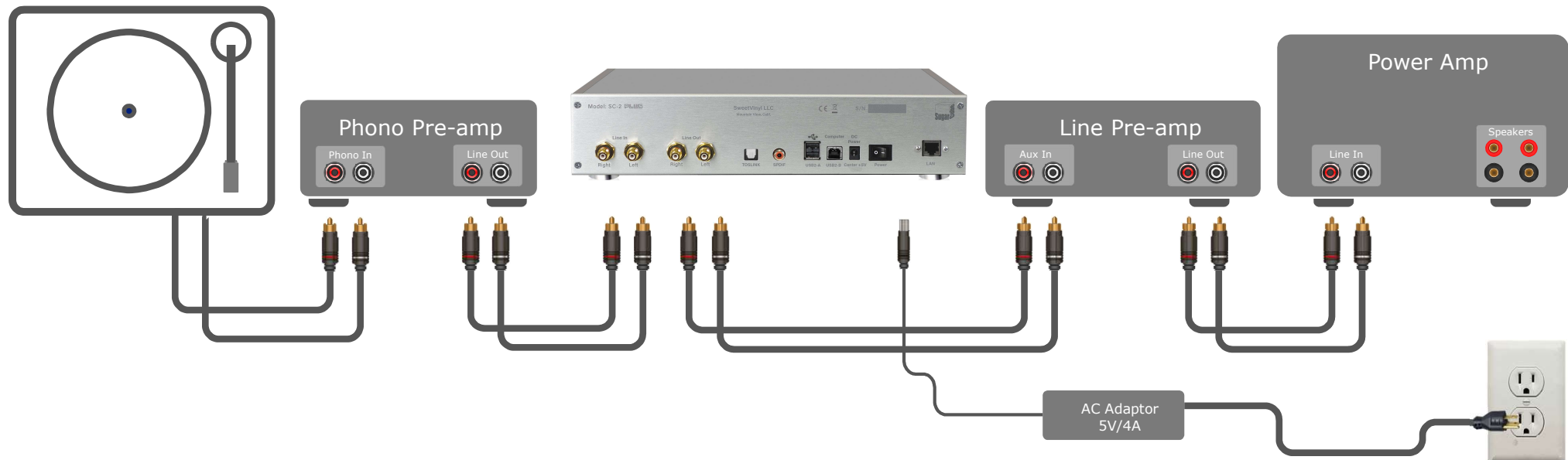
Section 2

Connecting SugarCube to Your Sound System

PLEASE BE SURE ALL
EQUIPMENT IS TURNED
OFF BEFORE ATTEMPTING
INSTALLATION



2.1 Connecting Your SugarCube



Connecting in series between a phono pre-amp and line-stage (pre-amp).

1. Plug your SugarCube power adapter into a power source.
2. Connect the output of your phono pre-amp to the **LINE IN** of the SC-2 Plus.
3. Connect the **LINE OUT** of the SC-2 Plus to an appropriate line-level input of your line-stage/pre-amp.

NOTE: DO NOT CONNECT THE **LINE OUT** OF THE **SUGARCUBE** TO THE PHONO INPUT ON YOUR EQUIPMENT.

2.2 Connecting Your SugarCube

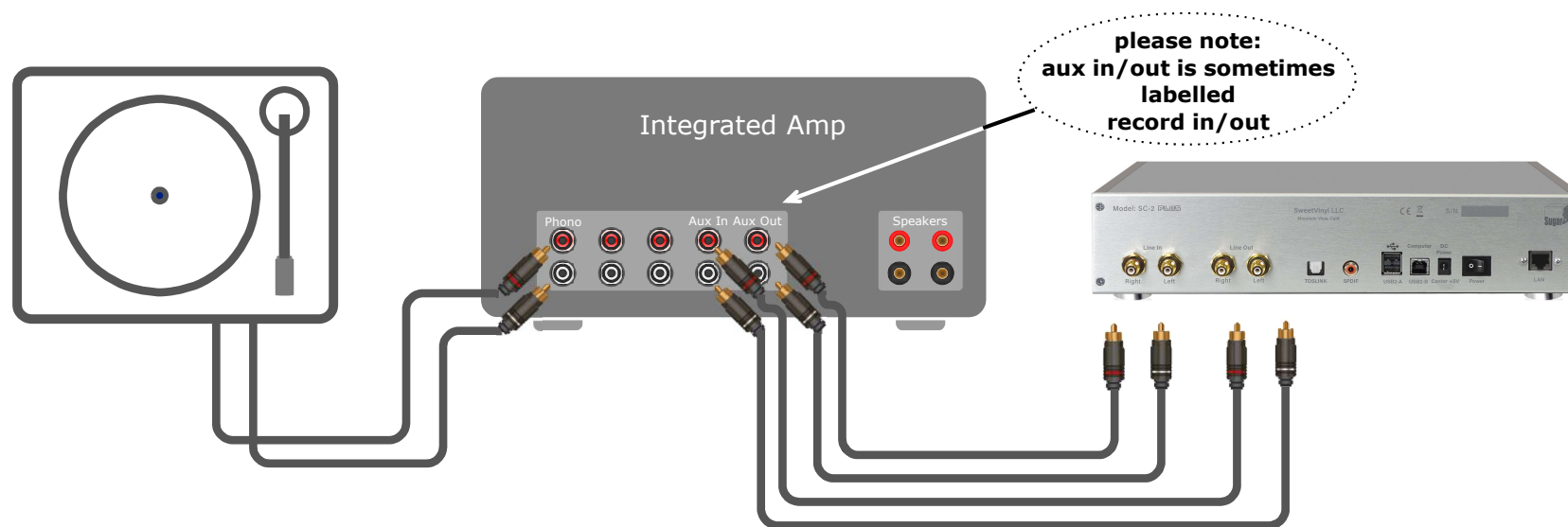


Connecting in series between a pre-amp, with phono input, and power amp

1. Connect the output of your pre-amp to the **LINE IN** of the SC2-Plus.
2. Connect the **LINE OUT** of the SC2-Plus to the input of your power amp.

2.3 Connecting Your SugarCube

Integrated Amp: using the 'tape loop'



Connecting to an integrated amplifier via Aux In/Out (or Rec In/Out).

1. Connect **AUX OUT** (or REC OUT) of your integrated amp (or processor) to the **LINE IN** of the SC2-Plus.
2. Connect **LINE OUT** of the SC2-Plus to **AUX IN** (or **REC IN**) of your integrated amp or processor.

NOTE: DO NOT CONNECT THE **LINE OUT** OF THE SUGARCUBE TO THE **PHONO INPUT** ON YOUR EQUIPMENT.

2.4 Connecting Your SugarCube

Integrated Amp: connecting between the preamp and power amp

Step 1: TURN OFF THE INTEGRATED AMP

Step 2: Wait a minute or two

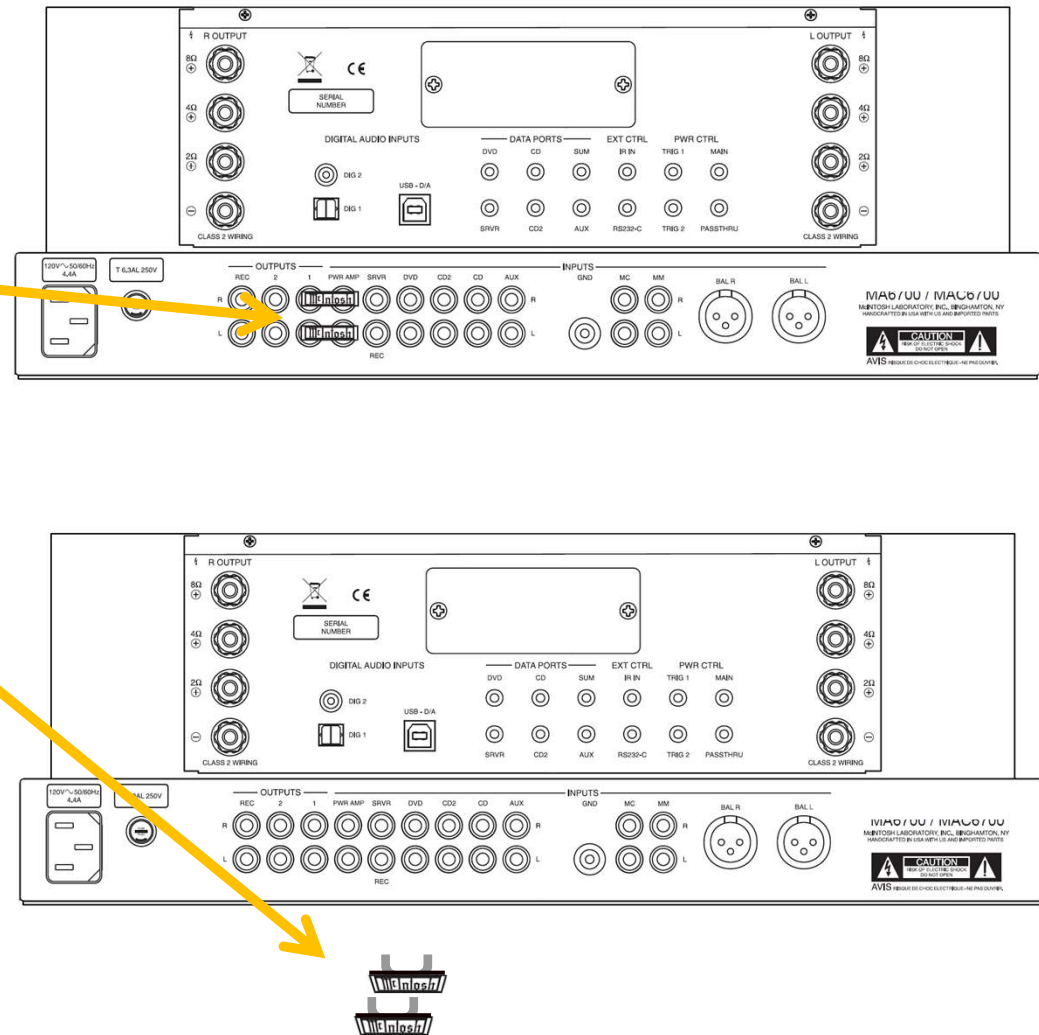
Step 3: locate the preamp-power amp jumpers

Step 4: DOUBLE-CHECK THAT THE AMP IS OFF

Step 5: Remove the jumpers

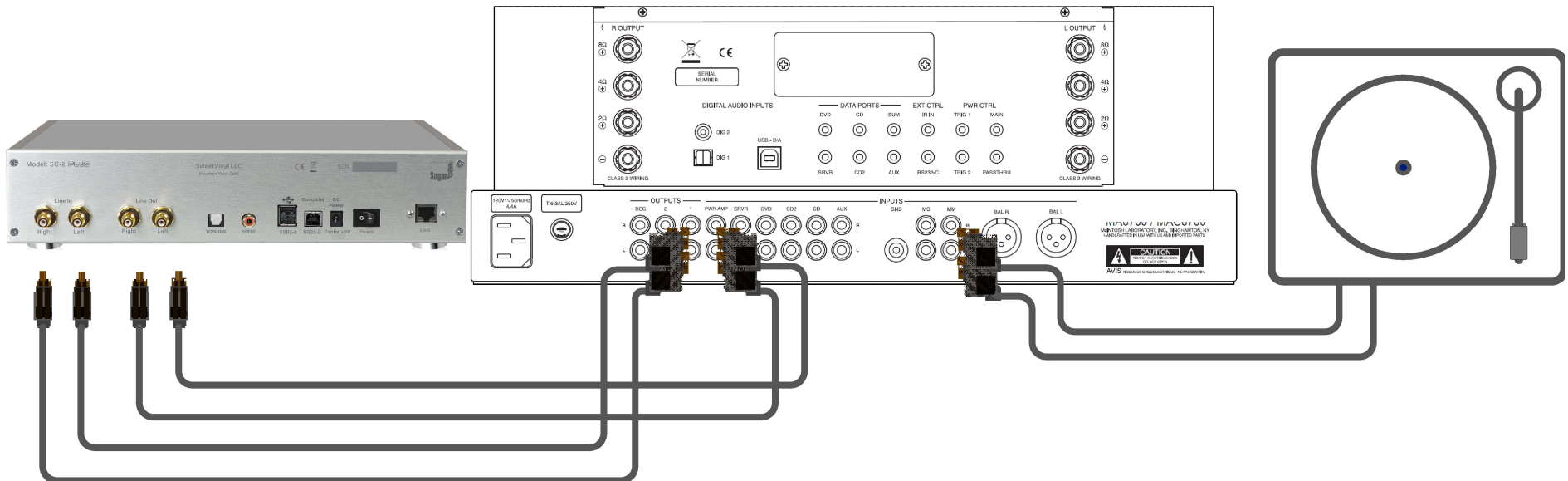
Step 6: Put the jumpers away in a safe place

Step 7: NEXT PAGE...



2.5 Connecting Your SugarCube

Integrated Amp: connecting between the preamp and power amp, cont.



Connecting to an integrated amplifier between the pre-amp and power-amp.

1. Connect **OUTPUT** (this is the pre-amp) of your integrated amp to the **LINE IN** of the SugarCube.
2. Connect **LINE OUT** of the SugarCube to **PWR AMP** (power amp input) of your integrated amp.
3. **ALWAYS TURN ON THE SUGARCUBE FIRST THEN THE AMP AND TURN OFF THE AMP FIRST THEN THE SUGARCUBE.**
4. If the integrated amp has a headphone output, it will likely have its own amplifier and may not be in the loop with the SugarCube. If that is the case, you will not hear the SugarCube in the headphones.
5. The SugarCube draws only about 11 watts, you can leave it on full time if you wish.
6. Not all integrated amps have a line out / power amp in or an Aux Out / In. If this is the case, you may need an external phono pre-amp.

2.6 Connecting Your SugarCube

Integrated Amp **WITHOUT** an AUX OUT/IN (or REC OUT/IN, etc.)



Many contemporary integrated amplifiers no longer have what was known as a 'tape loop' (back in the day of the cassette and reel-to-reel tape). Also, most modern home theatre processors that do have what are called 'effects loops' will **not** send audio from its phono input through the loop. In these cases, there is not a way to connect to SugarCube directly to these devices while using its built-in phono stage.

The solution is to add a phono stage (pre-amp) between your turntable and the SugarCube. Then connect the Line Out of the SugarCube to any available input on the integrated amp (except for the phono input, of course.)

2.7 Connecting Your SugarCube SC2-Plus

Connecting turntables with built-in phono pre-amps



For turntables with a built-in phono pre-amp, be sure to set the **PHONO/LINE** switch to **LINE** before plugging into the SugarCube.

Directly connecting the line-level output from your turntable.

1. Connect the turntable with a **built-in** phono pre-amp to **LINE IN**.
2. Be sure the turntable is set to **LINE** not **PHONO**.
3. Connect the **LINE OUT** of the SugarCube to any line-level input of your line stage, pre-amp, or integrated amp.

NOTE: SORRY TO BE A BROKEN RECORD, SO TO SPEAK, BUT DO NOT CONNECT THE **LINE OUT** OF THE SUGARCUBE TO THE **PHONO INPUT** ON YOUR EQUIPMENT.

2.8 Starting Up Your SugarCube



(1) SugarCube just powered on

(2) SugarCube is ready to go
(please expect about three minutes to ready.)

(3) Hold the 'Bypass' button for a few seconds to see LAN address (and WiFi SSID if using wireless.)

Please note:

When you power up your SugarCube, it will take about three minutes to complete its startup procedures. During this time, the LCD display will show snow (our humorous tip of the hat to the old days of analog TV). There is nothing wrong with your unit. We apologize for the wait but we have some seriously sophisticated software loading and it simply takes additional time. When you hear the results of our efforts, you will agree that it was worth the wait.

Section 3

Networking the SugarCube

- Part 1: Ethernet and WiFi
- Part 2: Chrome Browsers
- Part 3: Apple iOS
- Part 4: Android





3.1 Using Ethernet or WiFi with SugarCube

3.1.1 Pairing your SugarCube (when more than one is present)



If you own more than one SugarCube, while setting up the connection between the SugarCube and the mobile or web browser SugarCube app, push and release the '**Strength**' knob to obtain a unique pairing code.

3.1.2 Connecting Your SugarCube to a LAN or WiFi



If you would like to use the iOS or Android app or web browser to control your SugarCube SC2-Plus, please connect either the WiFi adapter supplied with your SugarCube or plug into a wired connection.

NOTES:

- ONLY BASIC OPERATIONS ARE FUNCTIONAL BY MEANS OF THE FRONT PANEL CONTROLS WITHOUT A CONNECT TO A NETWORK.

3.1.3 Connecting Your SugarCube via Wired Ethernet



- When connecting via wired Ethernet you need only open the iOS or Android App or follow a few steps to establish a browser connection.

3.1.4 Connecting Your SugarCube via WiFi



- If using the WiFi adapter, you will either use the iOS or Android App to setup WiFi, or use Ethernet and a browser to configure the WiFi parameters.



3.2 Access the SugarCube with a **Browser**

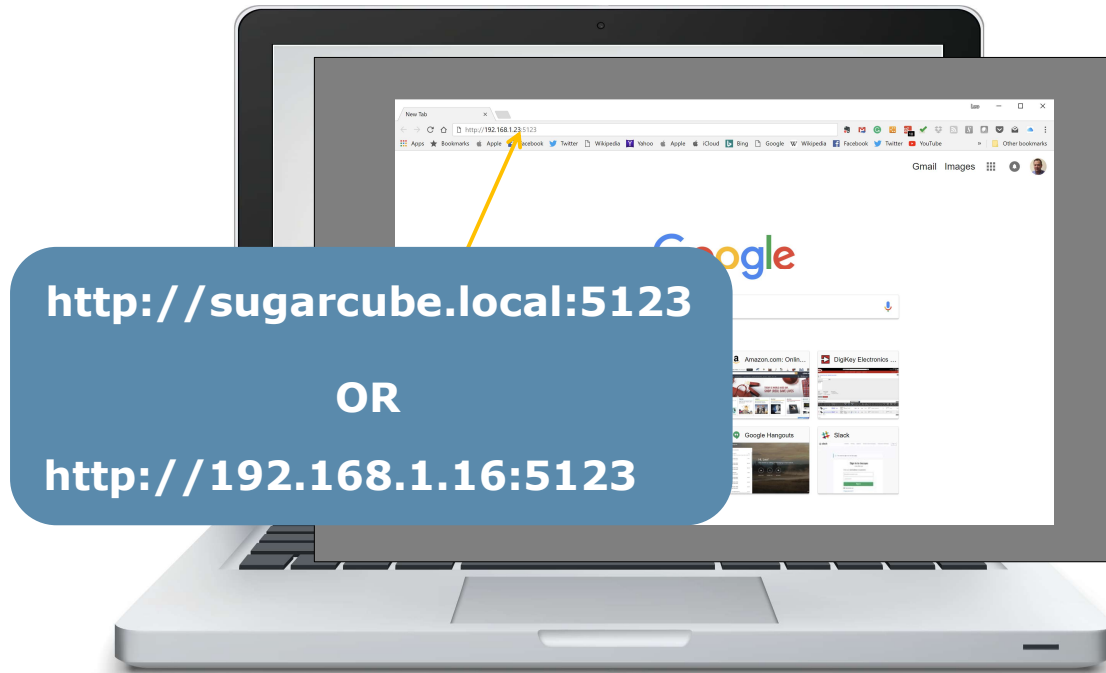
3.2.1 Controlling the SugarCube from a Web Browser

Finding the IP address of your SugarCube SC1-Plus



3.2.2 Controlling the SugarCube from a Web Browser

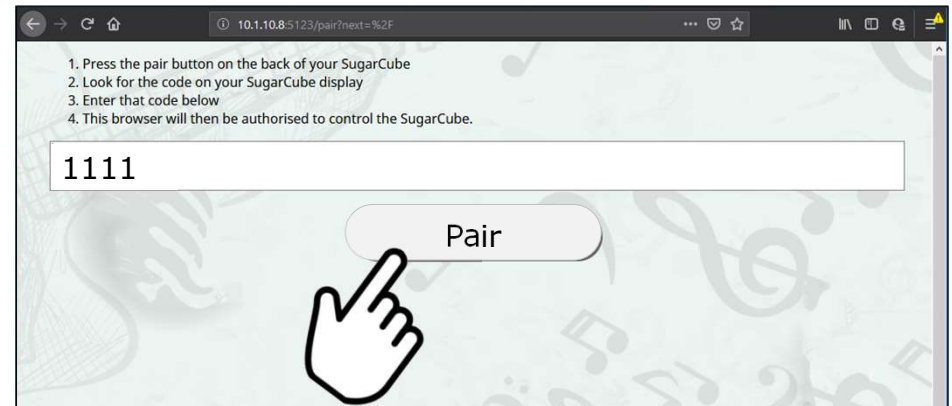
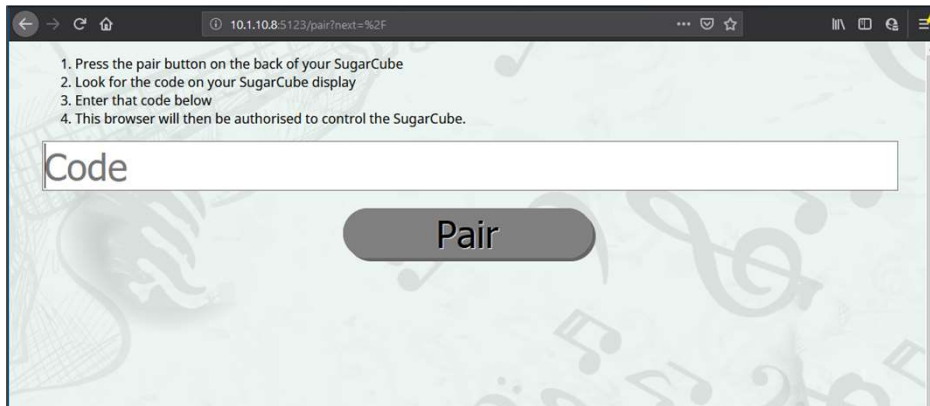
Entering either IP address (below) into the browser



- (1) Type ***http://sugarcube.local:5123*** to access your SugarCube.
- (2) If you own more than one SugarCube, enter the SugarCube's IP address into the **address bar** then add a colon ':' and '5123'.
- (3) Next, press '**Enter**' and the SugarCube user interface will appear in the browser.

Important Note: the SugarCube web interface is compatible with Google Chrome, Microsoft Edge, & Apple Safari web browsers. We cannot guarantee compatibility with any other brand.

3.2.3 Controlling the SugarCube from a Web Browser



NOTE: "1111" as a pairing code will be accepted on recent versions of the SugarCube software

1



3



(1) You will see a screen appearing in your browser to accept a pairing code.

2

(2) You don't need to do this, but you can press and release the knob on the front panel.

(3) Then, enter the four-digit pairing code displayed on the SugarCube display and click on 'Pair'.

3.2.4 Controlling the SugarCube from a Web Browser

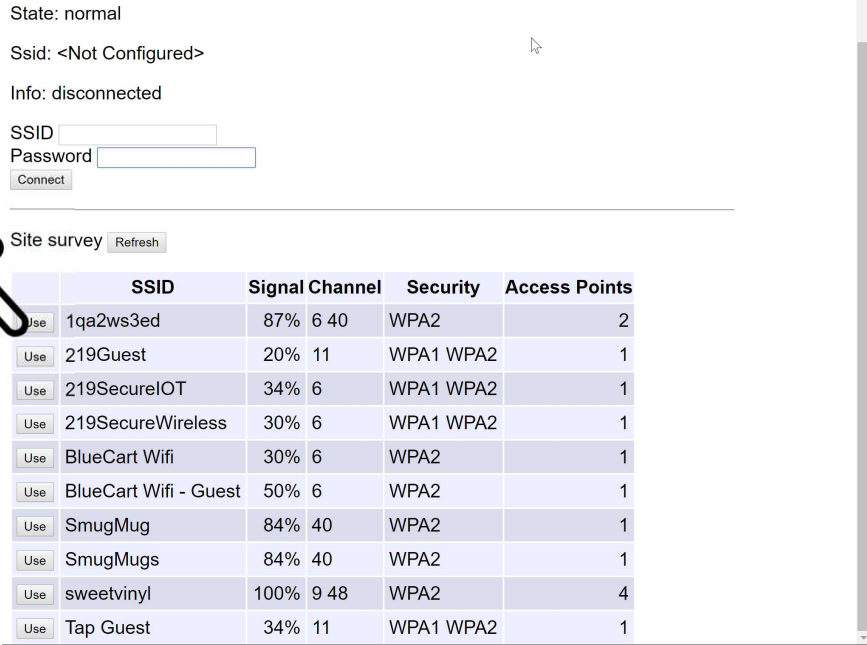
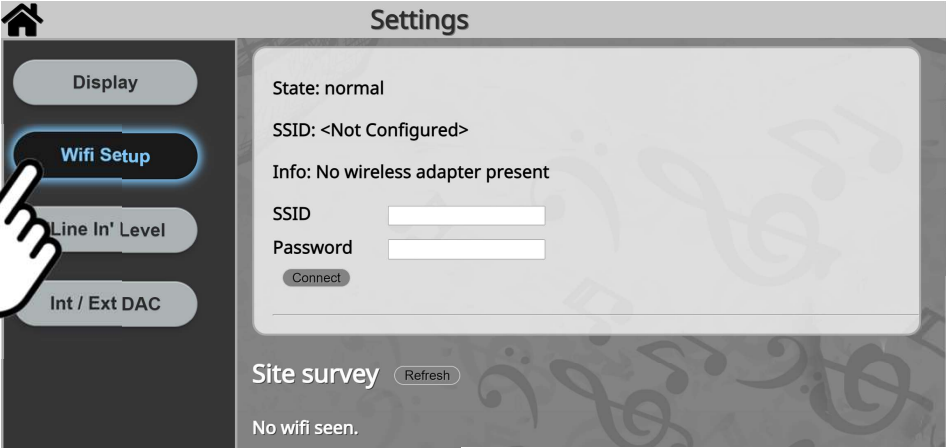


Important Note: the SugarCube web interface is compatible with Google Chrome, Microsoft Edge, Brave, & Apple Safari web browsers. We cannot guarantee compatibility with any other brand.

The SugarCube user interface will appear in the browser.

3.2.5 Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 1 of 4



	SSID	Signal	Channel	Security	Access Points
Use	1qa2ws3ed	87%	6 40	WPA2	2
Use	219Guest	20%	11	WPA1 WPA2	1
Use	219SecureIoT	34%	6	WPA1 WPA2	1
Use	219SecureWireless	30%	6	WPA1 WPA2	1
Use	BlueCart Wifi	30%	6	WPA2	1
Use	BlueCart Wifi - Guest	50%	6	WPA2	1
Use	SmugMug	84%	40	WPA2	1
Use	SmugMugs	84%	40	WPA2	1
Use	sweetviny1	100%	9 48	WPA2	4
Use	Tap Guest	34%	11	WPA1 WPA2	1

(1) If wired Ethernet is not available in your sound room, then take the SugarCube to your router just for these WiFi setup steps.

(2) You need both Ethernet and the WiFi stick plugged into the SugarCube.

(3) Find your WiFi name on the list of SSIDs.

3.2.6 Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 2 of 4

State: normal
Ssid: <Not Configured>
Info: disconnected
SSID: <your wifi>
Password:

Site survey

	SSID	Signal	Channel	Security	Access Points
Use	1qa2ws3ed	87%	6 40	WPA2	2
Use	219Guest	20%	11	WPA1 WPA2	1
Use	219SecureIoT	34%	6	WPA1 WPA2	1
Use	219SecureWireless	30%	6	WPA1 WPA2	1
Use	Wifi	30%	6	WPA2	1
Use	Wifi - Guest	50%	6	WPA2	1
Use		84%	40	WPA2	1
Use	SmugMugs	84%	40	WPA2	1
Use	sweetvinyl	100%	9 48	WPA2	4
Use	Tap Guest	34%	11	WPA1 WPA2	1

(4) Please select your network by clicking on the 'Use' button next to your SSID.

Inf disconnected
SSID: <your wifi>
Password: <password>

Site survey

	SSID	Signal	Channel	Security	Access
Use	1qa2ws3ed	80%	6 40	WPA2	
Use	219Guest	44%	11	WPA1 WPA2	
Use	219SecureIoT	30%	6	WPA1 WPA2	
Use	219SecureWireless	34%	6	WPA1 WPA2	
Use	BlueCart Wifi	34%	6	WPA2	
Use	SmugMug	87%	6 40	WPA2	

(5) Next, enter the password for your WiFi then click on 'Connect.'

3.2.7 Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 3 of 4

Press the pair button on your SugarCube for at least 5 seconds to start the setup wifi network.

State: connect

Ssid: s1 <your wifi>

Info: connecting (getting IP configuration)

SSID

Password

Site survey

	SSID	Signal	Channel	Security	Access Points
Use	1qa2ws3ed	87%	6 40	WPA2	2
Use	219Guest	20%	11	WPA1 WPA2	1
Use	219SecureIoT	34%	6	WPA1 WPA2	1
Use	219SecureWireless	30%	6	WPA1 WPA2	1
Use	BlueCart Wifi	30%	6	WPA2	1
Use	BlueCart Wifi - Guest	50%	6	WPA2	1
Use	SmugMug	84%	40	WPA2	1
Use	SmugMugs	84%	40	WPA2	1

WiFi IP address

Your WiFi SSID

Wired IP address



(6) The system will indicate that it is connecting to your WiFi.

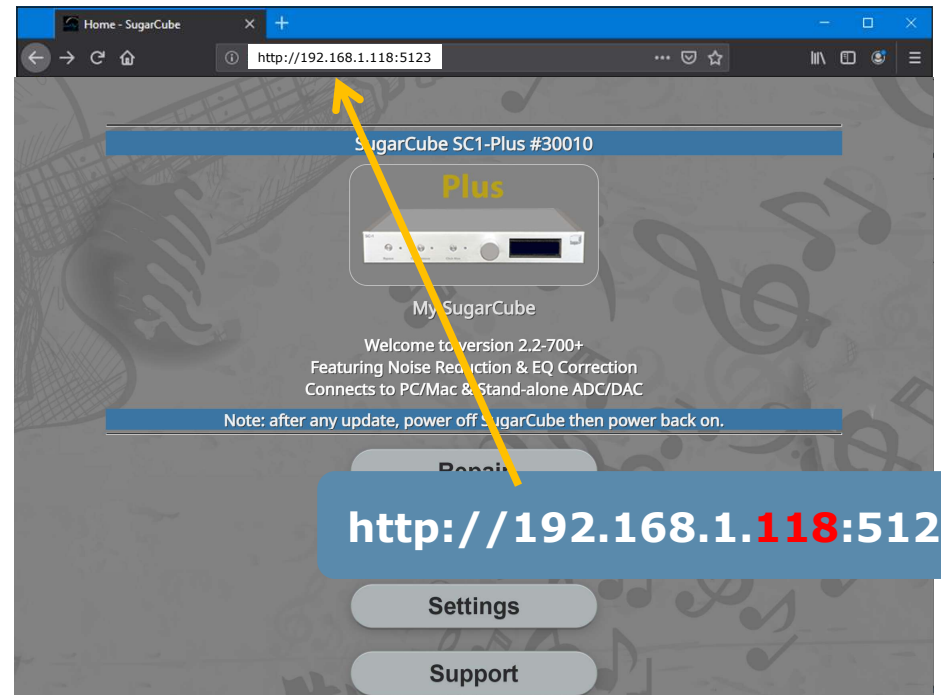
(7) Once connected, you will now see, scrolling by, both your new WiFi address (w) and your currently wired Ethernet IP address (e). Remember, you are currently communicating with the SugarCube via the wired IP address.

3.2.8 Controlling the SugarCube from a Web Browser

Setting up WiFi (in place of wired Ethernet), 4 of 4



(9) So, when you disconnect the wire, your browser will no longer communicate. You will have to change the IP address in your browser, in this example, to the WiFi address ending in '.118'. (Once the wire is disconnected, the front panel display will show only the WiFi IP address.)



(10) Usually, you need only to change the last group of numbers and press 'Enter', and you should reconnect (sometimes you may have to re-enter the entire address).

(11) Please note, if your SugarCube cannot connect to your WiFi, contact support@sweetvinyl.com with the details of your system (WiFi make and model and the serial number of your SugarCube.)

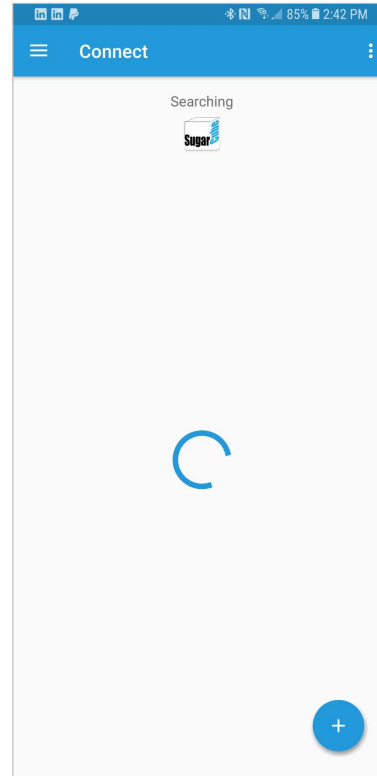


3.3 Using **iOS App** with SugarCube

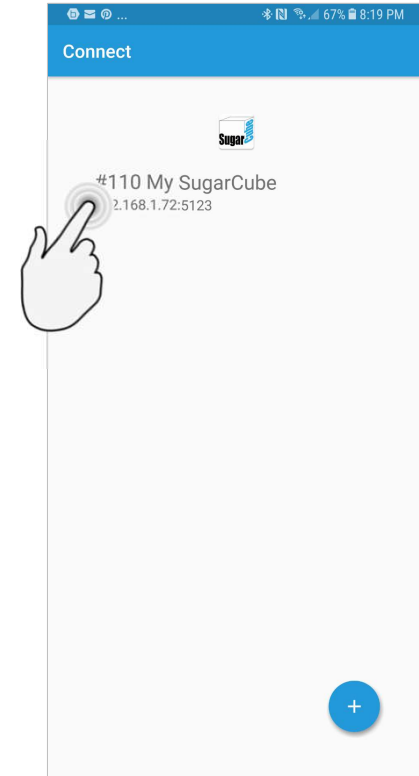
3.3.1 Connecting App with SugarCube on **Wired Ethernet**



(1) App is loading

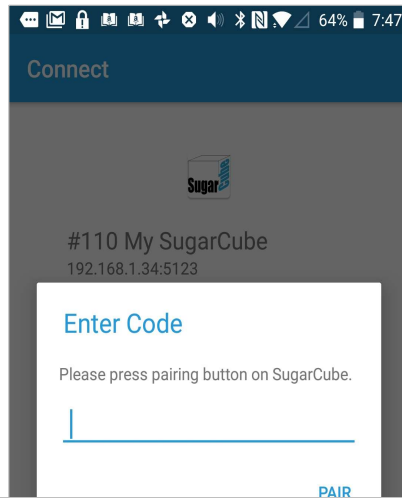


(2) App is searching for SugarCubes on the LAN



(3) App has found a SugarCube and will now initiate a connection.

3.3.2 Connecting App with SugarCube on **Wired Ethernet**

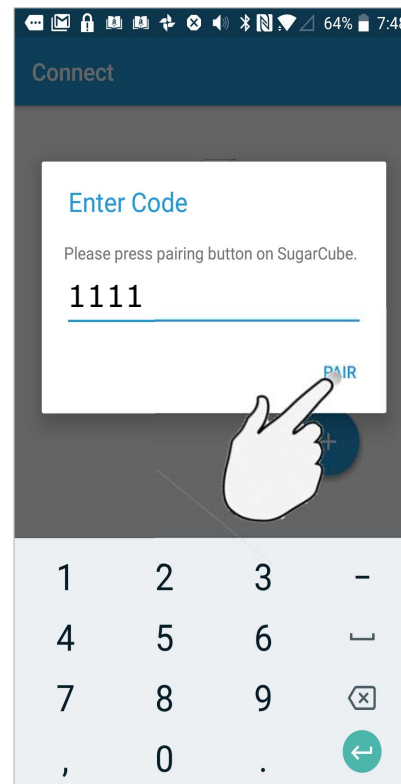


For pairing, enter:

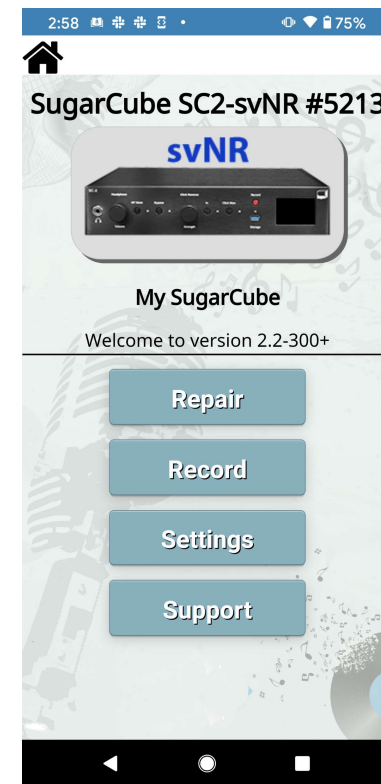
1 1 1 1

**as a pairing code
(you may still get a code
from your SugarCube,
if you wish).**

(4) The app needs you to enter a pairing code, press the knob on the front panel.

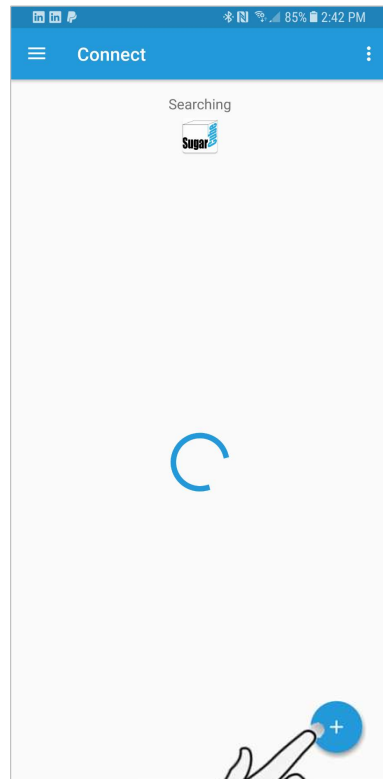


(5) App needs you to enter the pair code '1111' or what is displayed on the front panel display then touch 'PAIR'.

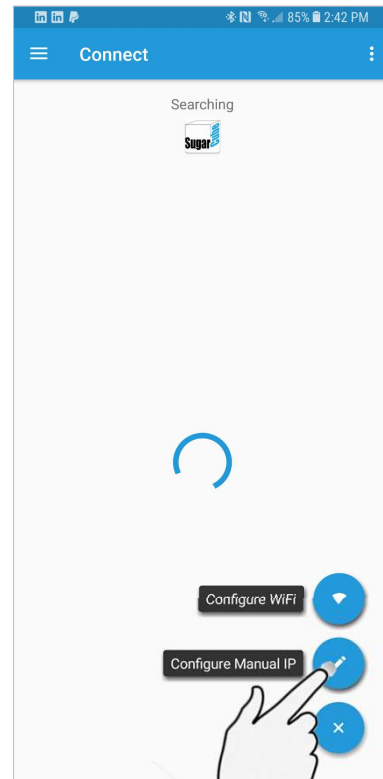


(6) App is linked to SugarCube and ready for use.

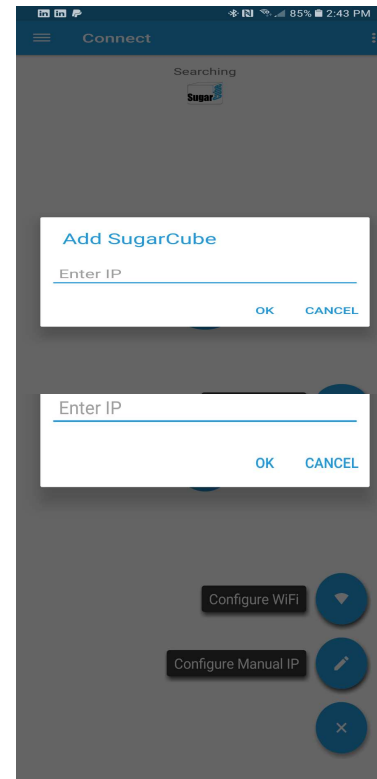
3.3.3 Connecting App with SugarCube Using IP Address



(1) If, for any reason, you wish to manually connect to your SugarCube touch the menu button in the lower right of the screen.



(2) Select 'Configure Manual IP'.

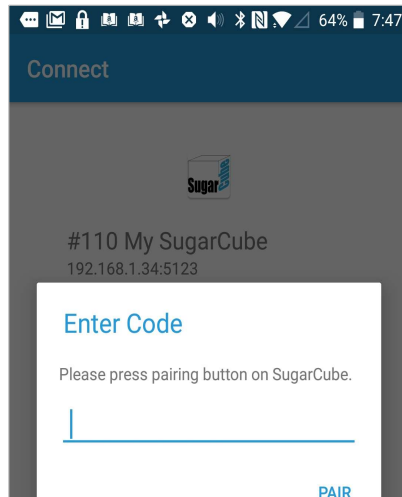


(3) You will then see a window to enter the IP address of the SugarCube. This address can be seen by pressing and holding 'Bypass' for a few seconds.



(4) Enter the IP address as displayed on the front panel display.

3.3.4 Connecting App with SugarCube Using IP Address

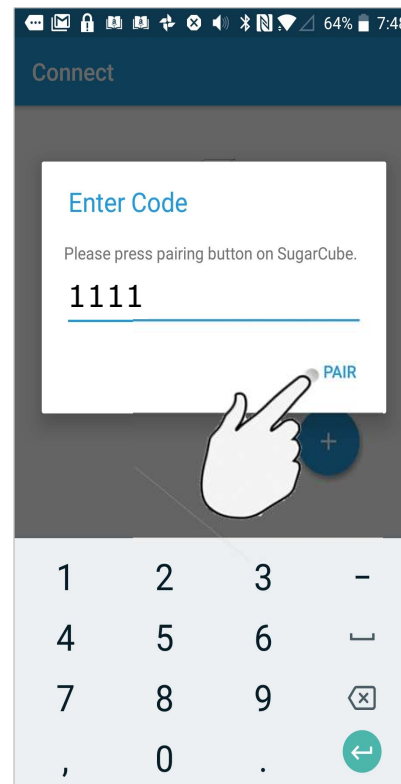


For pairing, enter:

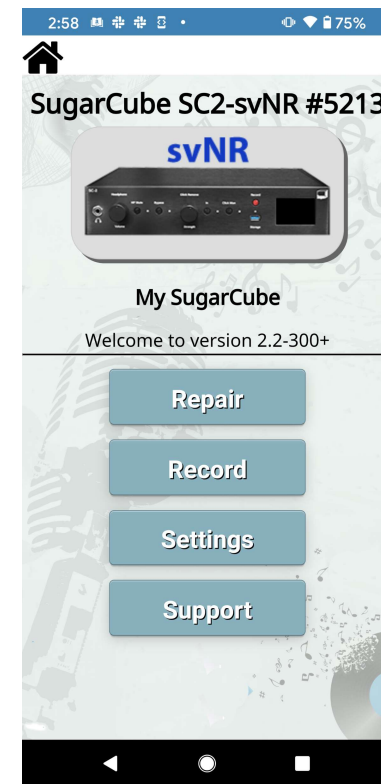
1 1 1 1

**as a pairing code
(you may still get a code
from your SugarCube,
if you wish).**

(5) The app need you to press the knob on the front panel.



(6) App needs you to enter the pair code displayed on the front panel display then touch 'PAIR'.

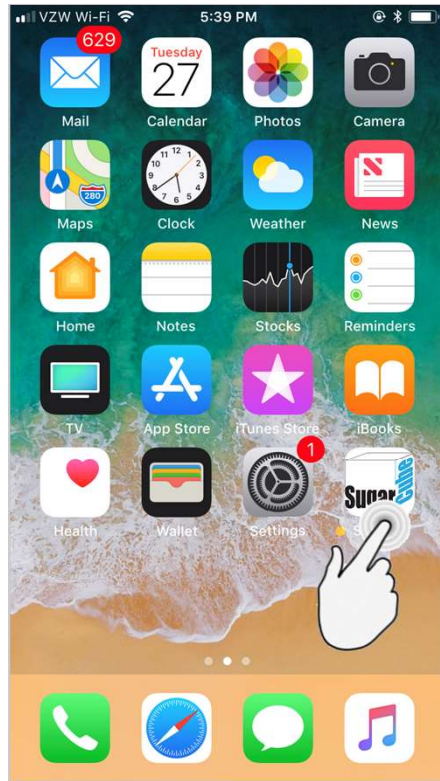


(7) App is linked to SugarCube and ready for use.

3.3.5 Setting up iOS device to control SugarCube with **WiFi**

1. iPhone, iPad, Touch.
2. Network with WiFi your Apple device can communicate with and that can be connected to your SugarCube with WiFi or wired Ethernet.
3. From the Apple App Store, you will be able to download the 'SugarCube' app to your Apple mobile device. (Look for: SugarCube by SweetVinyl.)
4. The WiFi dongle supplied with your SugarCube.

3.3.6 Connecting App with SugarCube on WiFi



(1) Launch the SugarCube app



(2) App is loading

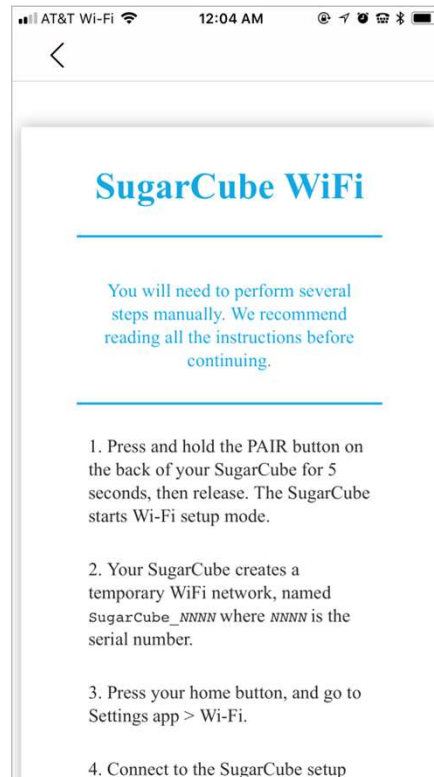


(3) App is searching for SugarCubes already setup on the LAN if any. For the first time user, nothing will be found, of course.

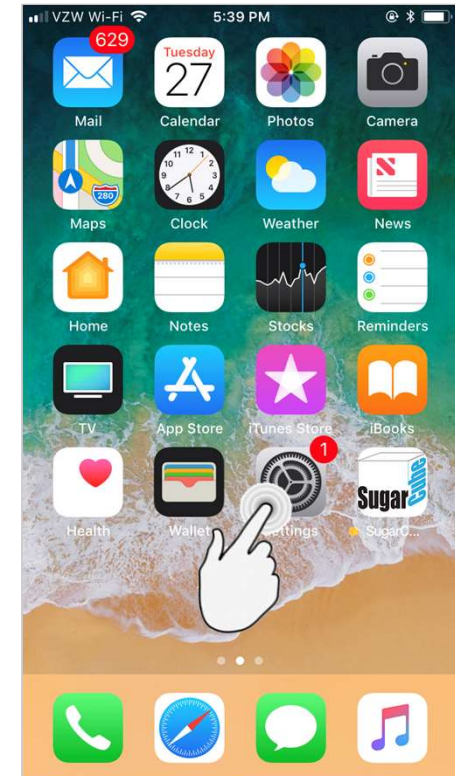
3.3.7 Connecting App with SugarCube on WiFi



(4) No SugarCube is found so the app offers the option to run WiFi setup.

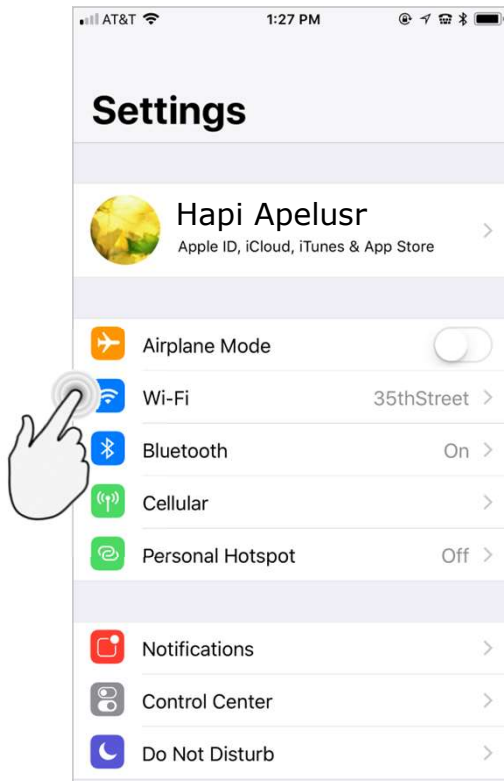


(5) The user is instructed to press and hold the front panel knob for 5 seconds to put the SugarCube into 'WiFi Setup Mode' which will be displayed on the front panel.

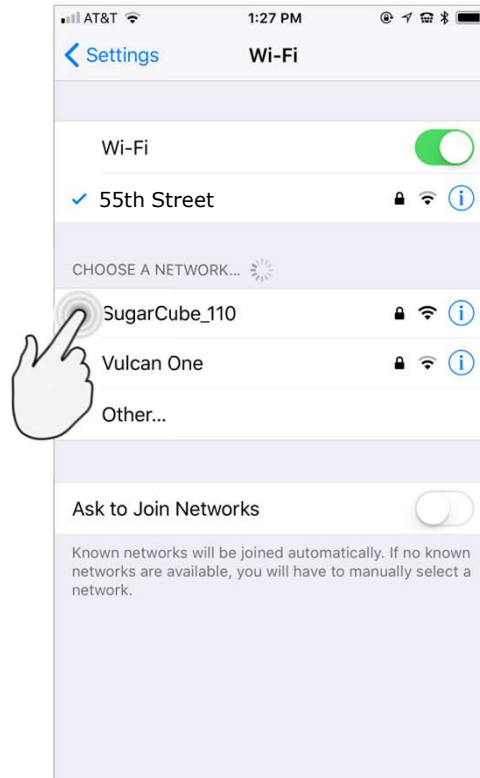


(6) The instructions direct the user to press the 'Home' key to get back to the 'Home' screen and then launch 'Setup'.

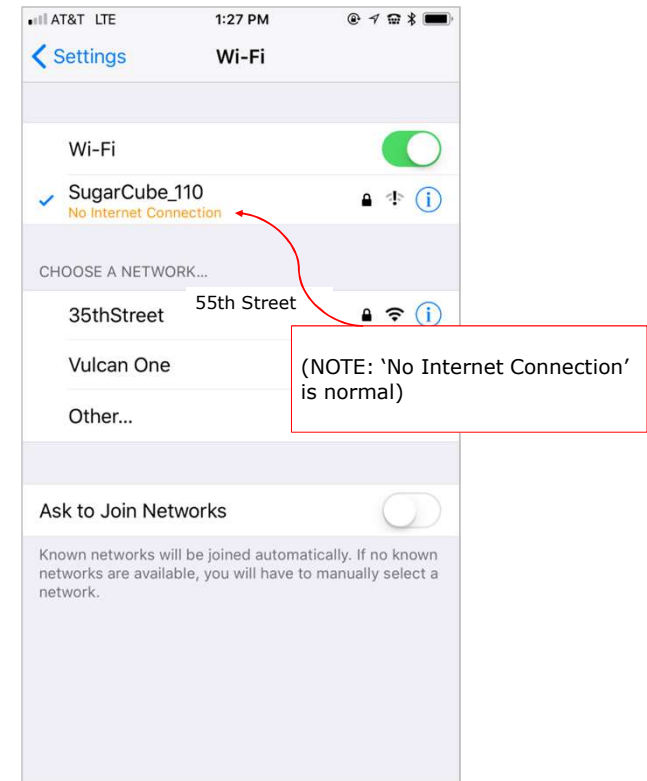
3.3.8 Connecting App with SugarCube on WiFi



(7) When in 'Settings', select 'WiFi'

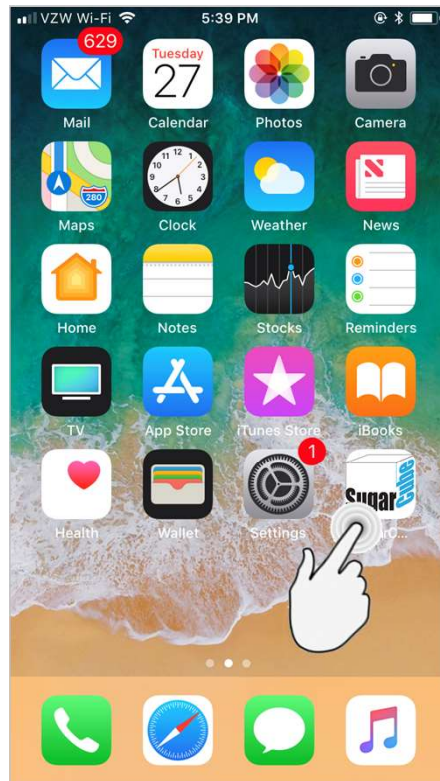


(8) Select 'SugarCube_nnnn', (whatever number appears after 'SugarCube_' is your device's serial number)

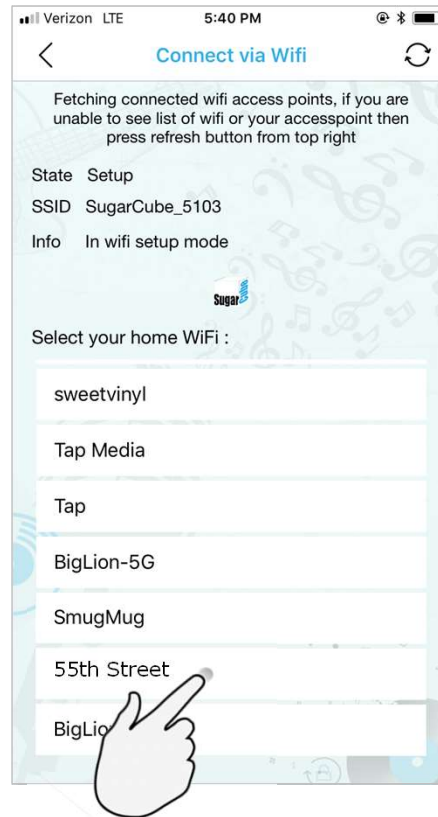


(9) With the 'SugarCube_nnnn' selected, your SugarCube is now temporarily acting as an access point.

3.3.9 Connecting App with SugarCube on WiFi



(10) Relaunch the SugarCube app



(11) Select the WiFi network that you wish to use



(12) Enter the WiFi password for your network

3.3.10 Connecting App with SugarCube on WiFi

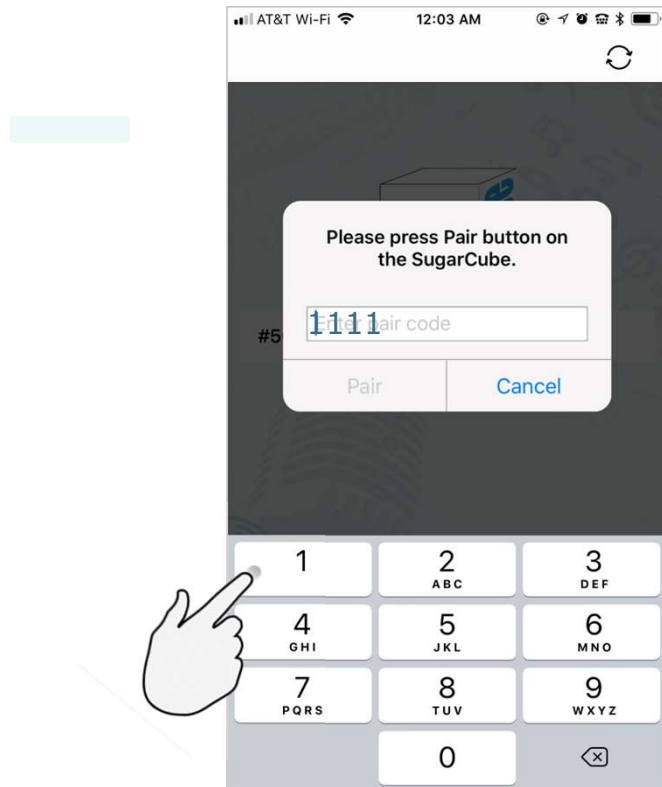


(13) The app will now complete the connection to your WiFi network and also do a little housekeeping.



(14) The app will then display the SugarCube(s) found on your network. Select the appropriate device.

3.3.11 Connecting App with SugarCube on WiFi



(15) When asked to 'Pair', enter '1111' (or you can press the front panel knob of the SugarCube and read the code from the display).



(16) You will then see the 'Control' screen and you are done, congratulations!

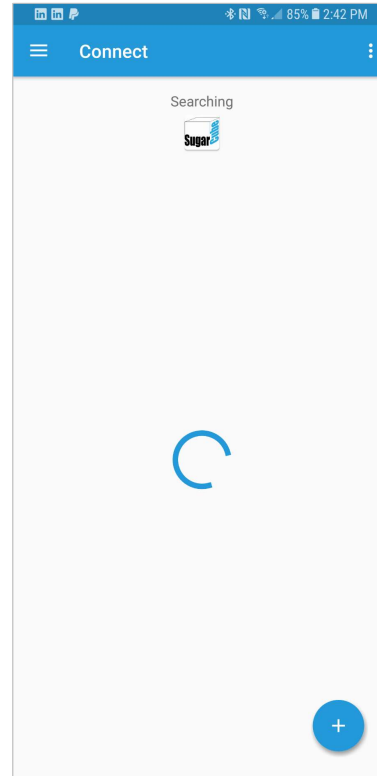


3.4 Using **Android App** with SugarCube

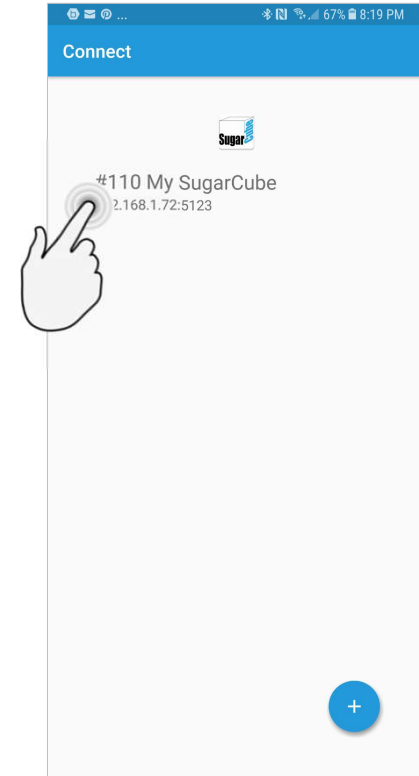
3.4.1 Connecting App with SugarCube on **Wired Ethernet**



(1) App is loading

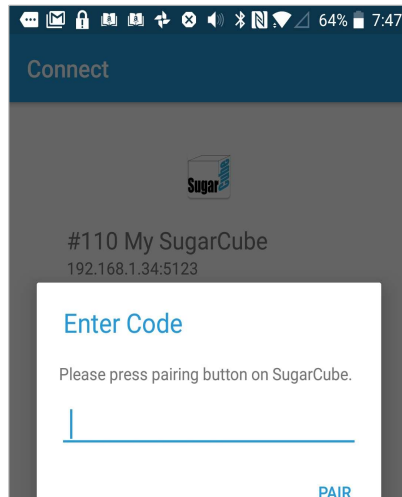


(2) App is searching for SugarCubes on the LAN



(3) App has found a SugarCube and will now initiate a connection.

3.4.2 Connecting App with SugarCube on **Wired Ethernet**

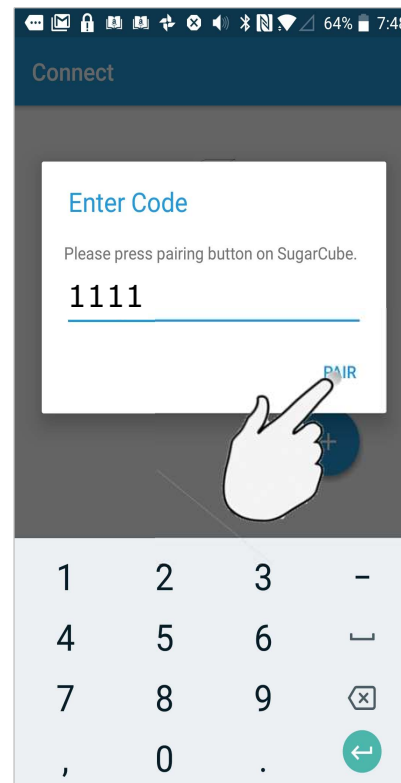


For pairing, enter:

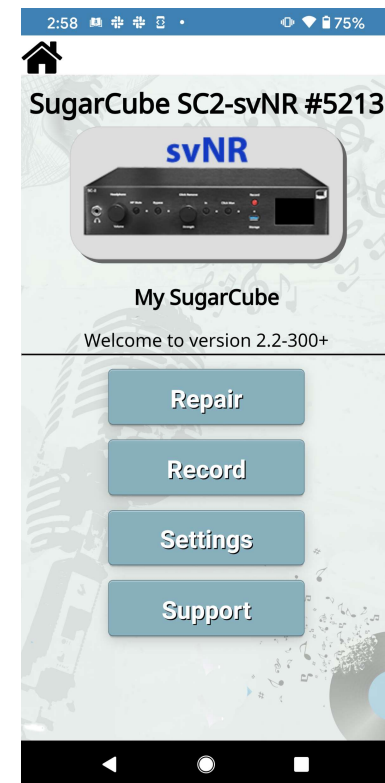
1 1 1 1

**as a pairing code
(you may still get a code
from your SugarCube,
if you wish).**

(4) The app needs you to enter a pairing code, press the knob on the front panel.

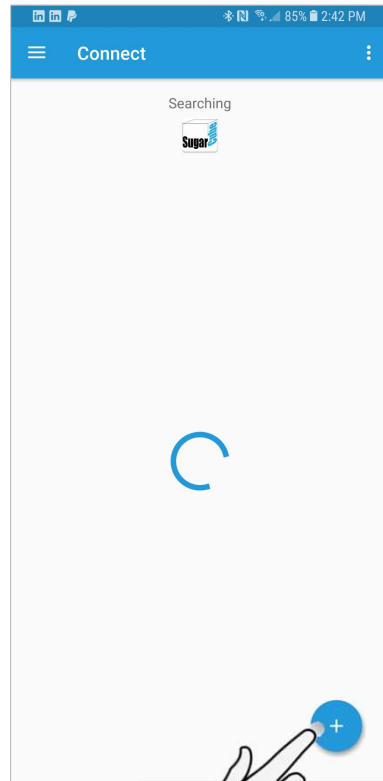


(5) App needs you to enter the pair code '1111' or what is displayed on the front panel display then touch 'PAIR'.

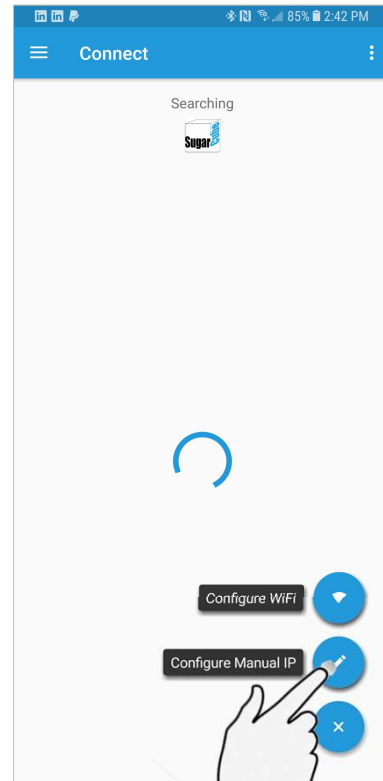


(6) App is linked to SugarCube and ready for use.

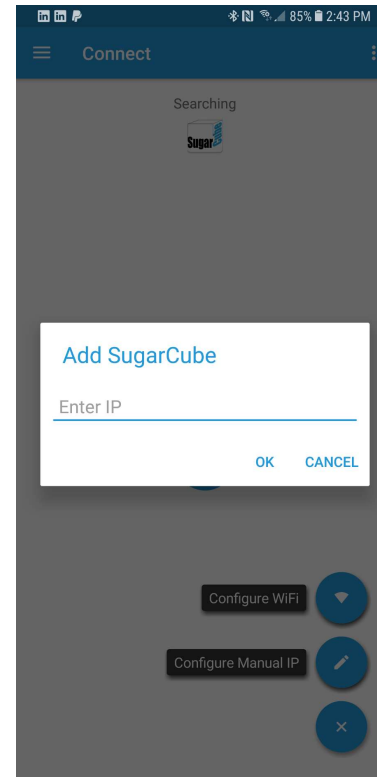
3.4.3 Connecting App with SugarCube Using IP Address



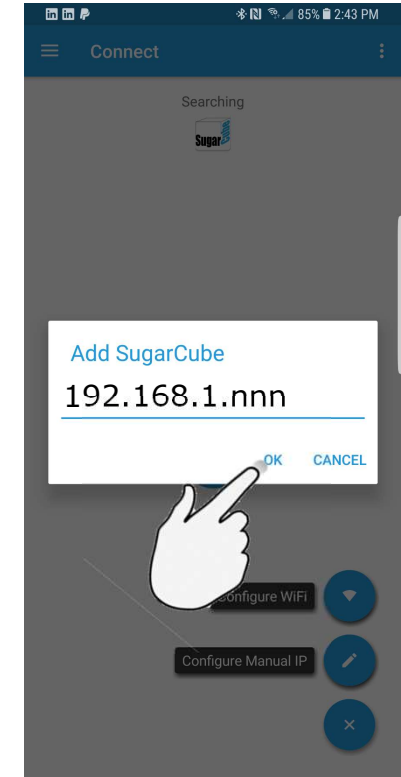
(1) If, for any reason, you wish to manually connect to your SugarCube touch the menu button in the lower right of the screen.



(2) Select 'Configure Manual IP'.

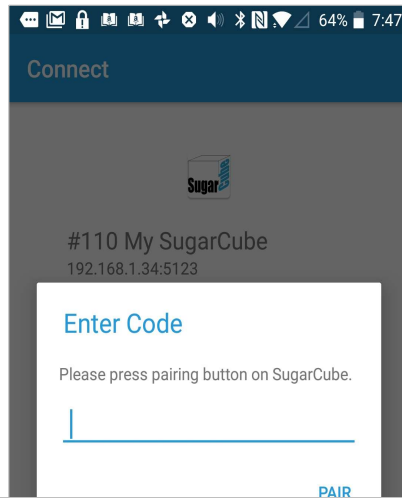


(3) You will then see a window to enter the IP address of the SugarCube. This address can be seen by pressing and holding 'Bypass' for a few seconds.



(4) Enter the IP address as displayed on the front panel display.

3.4.4 Connecting App with SugarCube Using IP Address

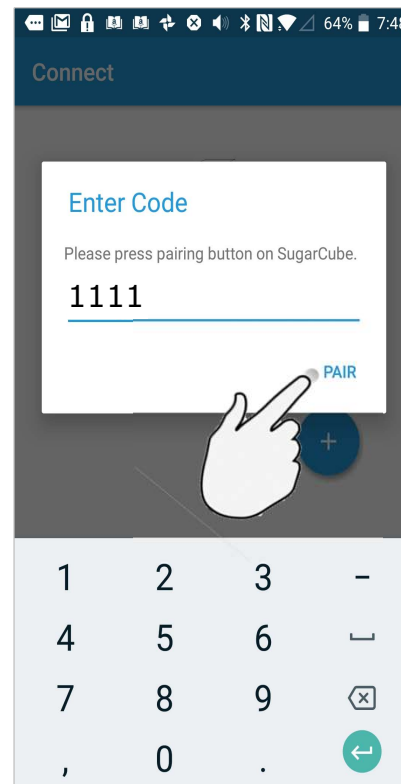


For pairing, enter:

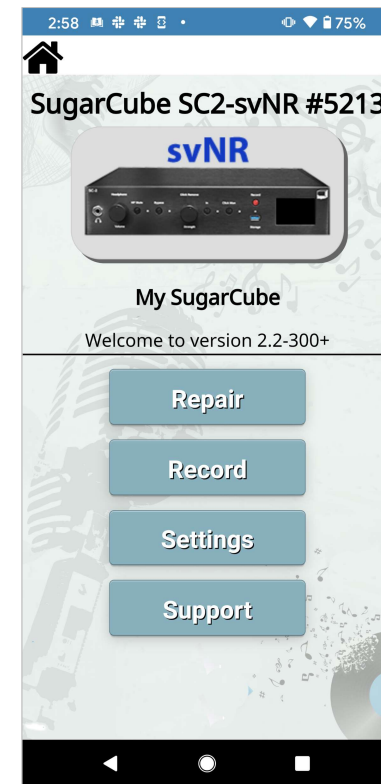
1 1 1 1

**as a pairing code
(you may still get a code
from your SugarCube,
if you wish).**

(5) The app need you to press the knob on the front panel.



(6) App needs you to enter the pair code displayed on the front panel display then touch 'PAIR'.



(7) App is linked to SugarCube and ready for use.

3.4.5 Setting up Android to control SugarCube with **WiFi**

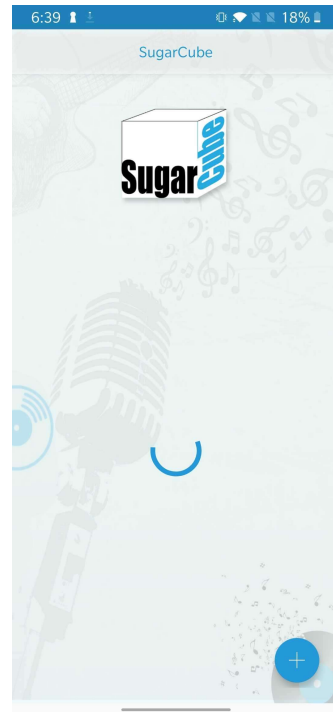
You will need:

1. An Android phone or tablet.
2. To connect with WiFi on your Android device on the same network that you plan to use with your SugarCube with WiFi or wired Ethernet.
3. Download the Android App from the Google Play Store.
(Look for: SugarCube by SweetVinyl.)
4. To plug in the WiFi dongle supplied with your SugarCube or a wired Ethernet cable from a port on your router.

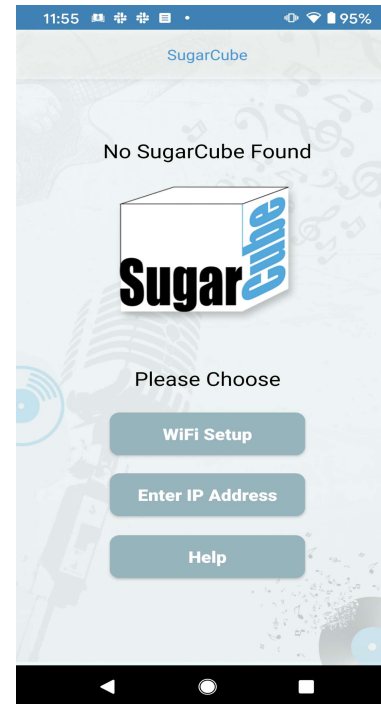
3.4.6 Connecting App with SugarCube on WiFi



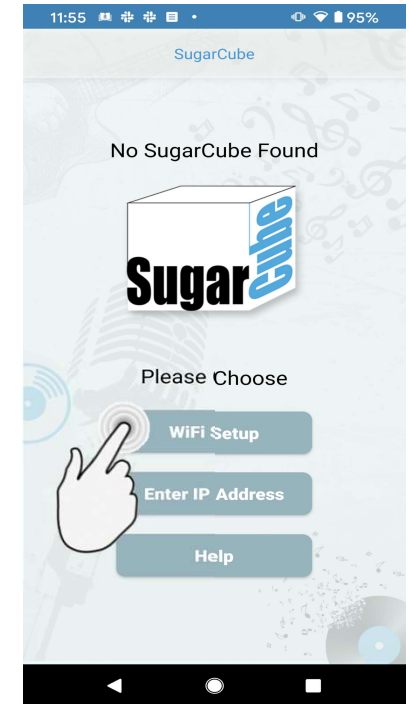
(1) App is loading



(2) App is searching for SugarCubes on the LAN

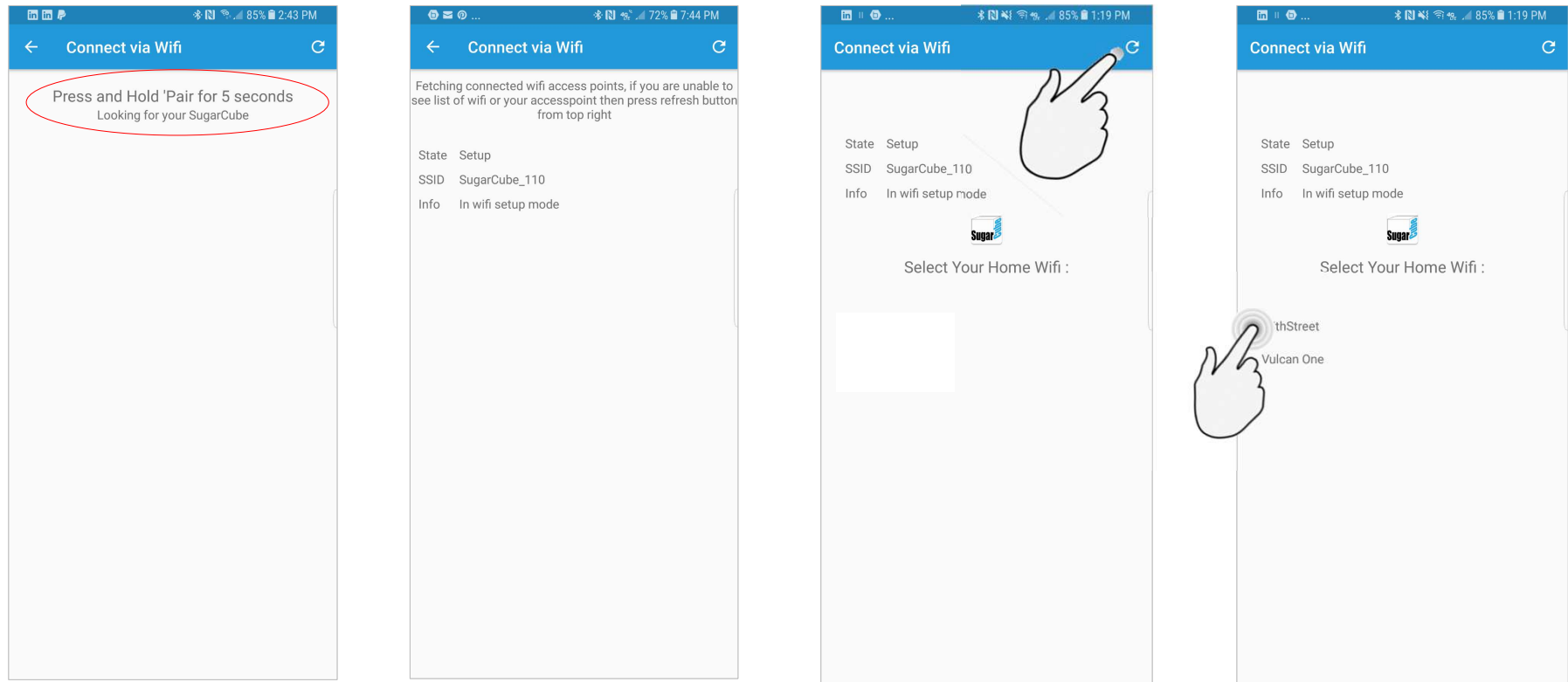


(3) No SugarCube is found so the app offers the option to run WiFi setup.



(4) User taps 'WiFi Setup' to start the process.

3.4.7 Connecting App with SugarCube on WiFi



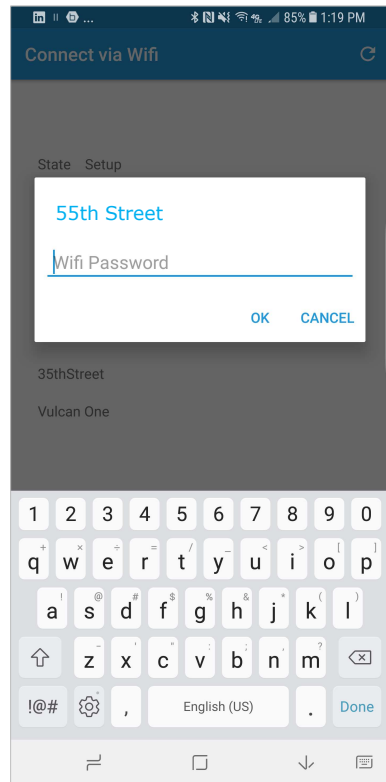
(5) Next, press and hold the pairing button (on the rear panel) for 5 seconds as instructed.

(6) After this, your SugarCube will be configured as an access point and display its SSID (e.g. SugarCube_110) In this screen the app has now connected to it automatically.

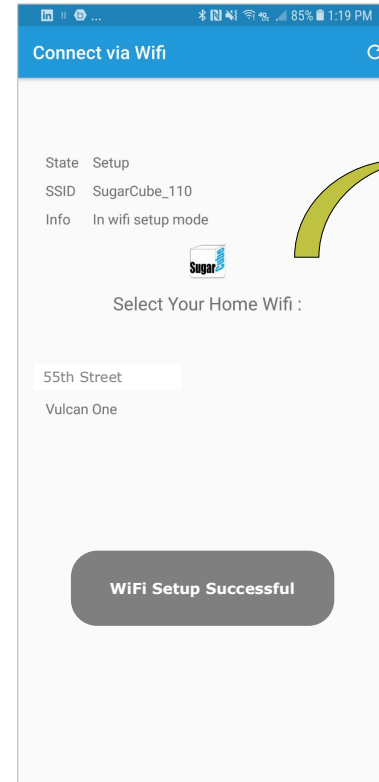
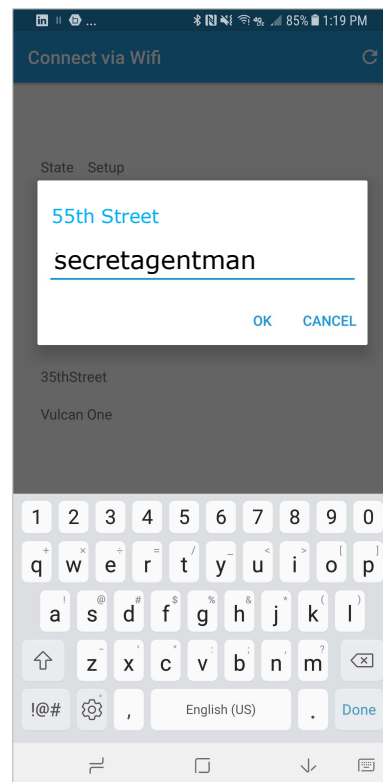
(7) Next, the SugarCube will find your available WiFi connections. This could take a minute but if nothing shows, please press the refresh (upper right).

(8) When you see your network, select the desired connection.

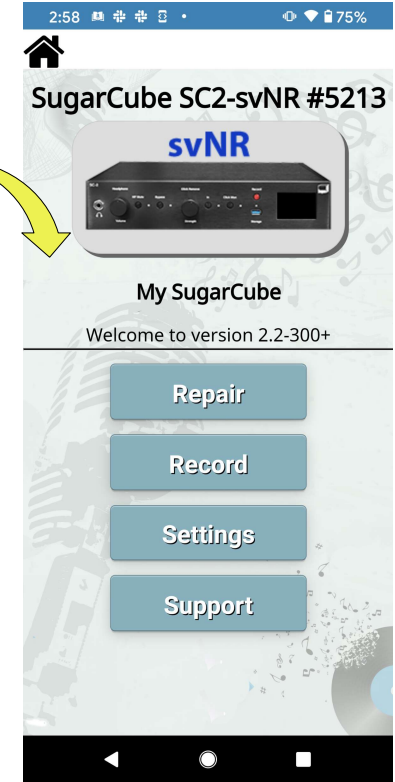
3.4.8 Connecting App with SugarCube on WiFi



(9) Next, enter the password for your WiFi network.



(10) The app has connected successfully with the SugarCube and will now jump to the main screen.



(11) The connection process is complete and you are ready to go.

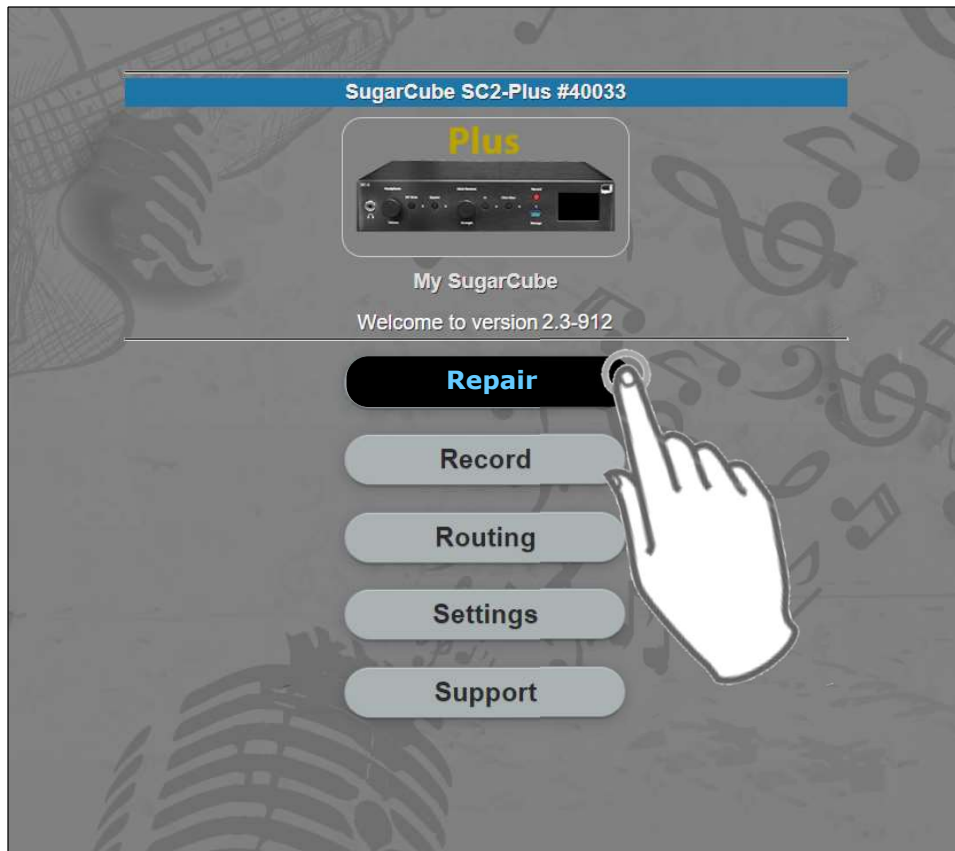
Section 4

Controlling SugarCube via a browser or app

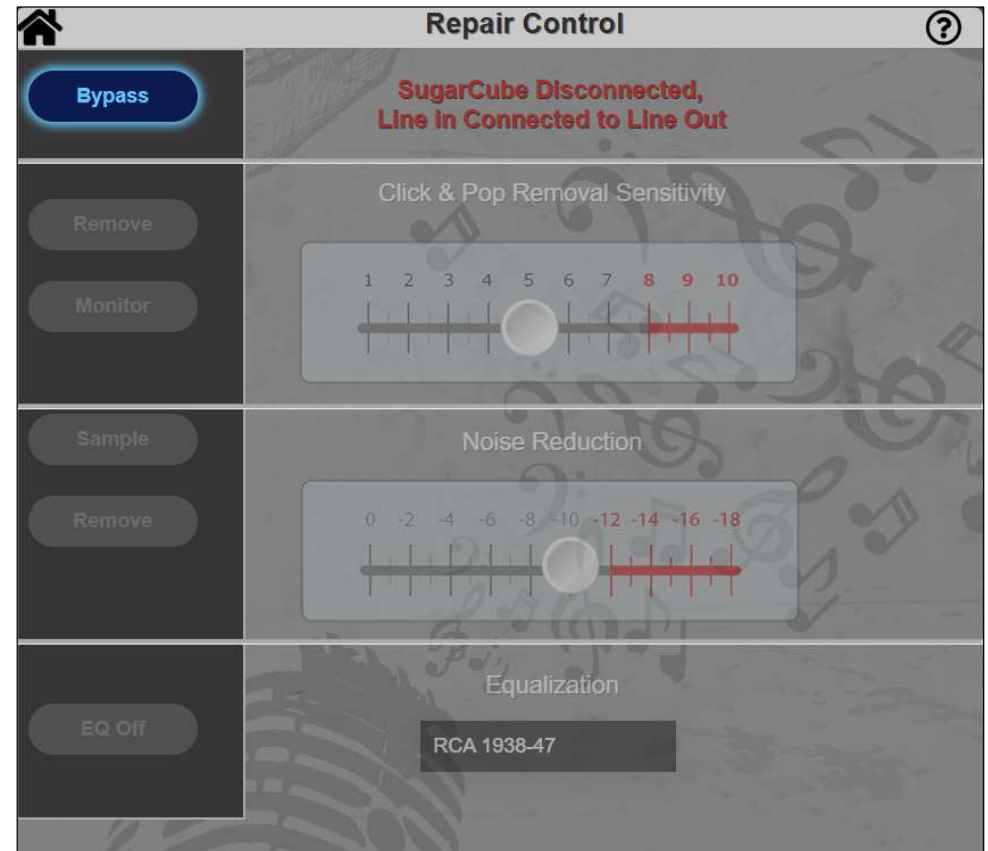
Chrome, Edge, or Safari
browser &
iOS or Android app



4.1 Home Page and Repair Control



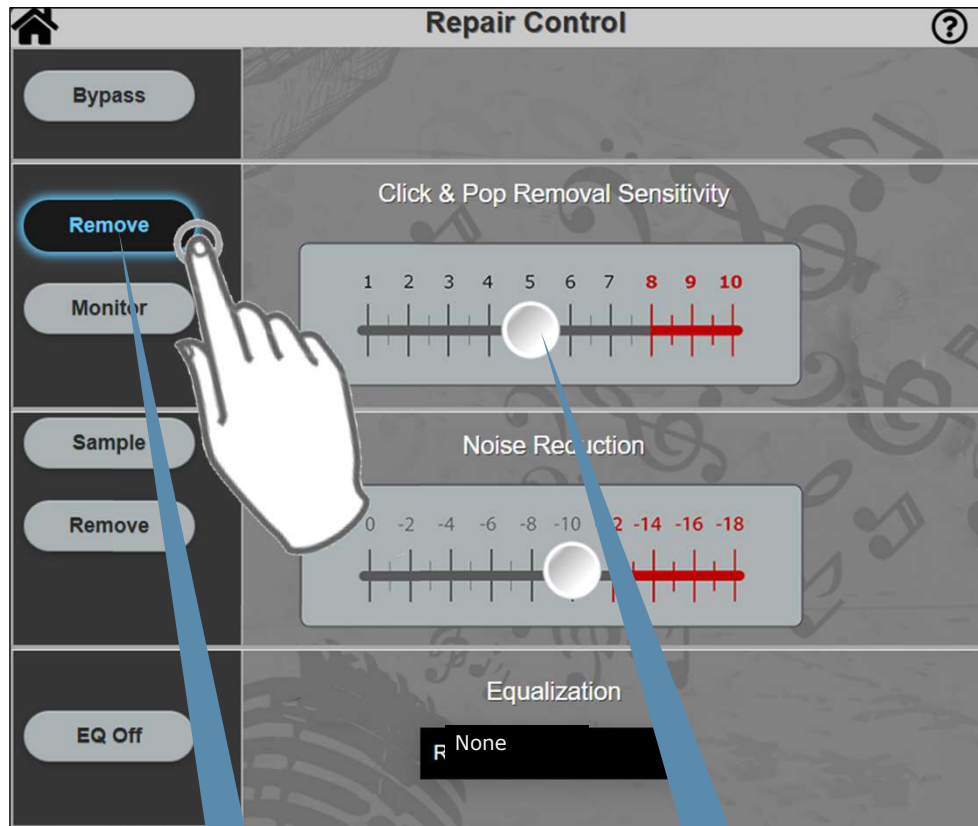
(1) Click on 'Repair'



(2) Click and Pop and Bypass functions available mirror the front panel controls. A change here is reflected on the front panel and a change on the front panel changes the above.

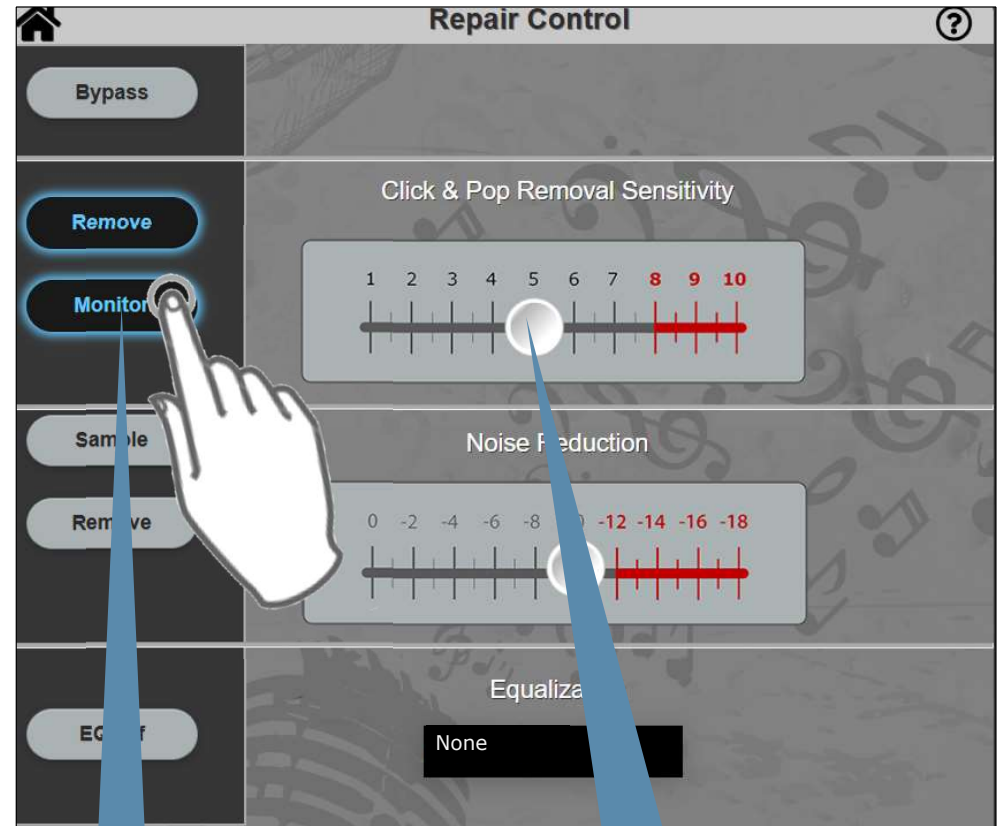
4.2 Repair Control Functions

Controlling Click and Pop Removal



Adjust the slider to listen to the result of setting 'Strength'.

The Click and Pop removal function has been engaged.

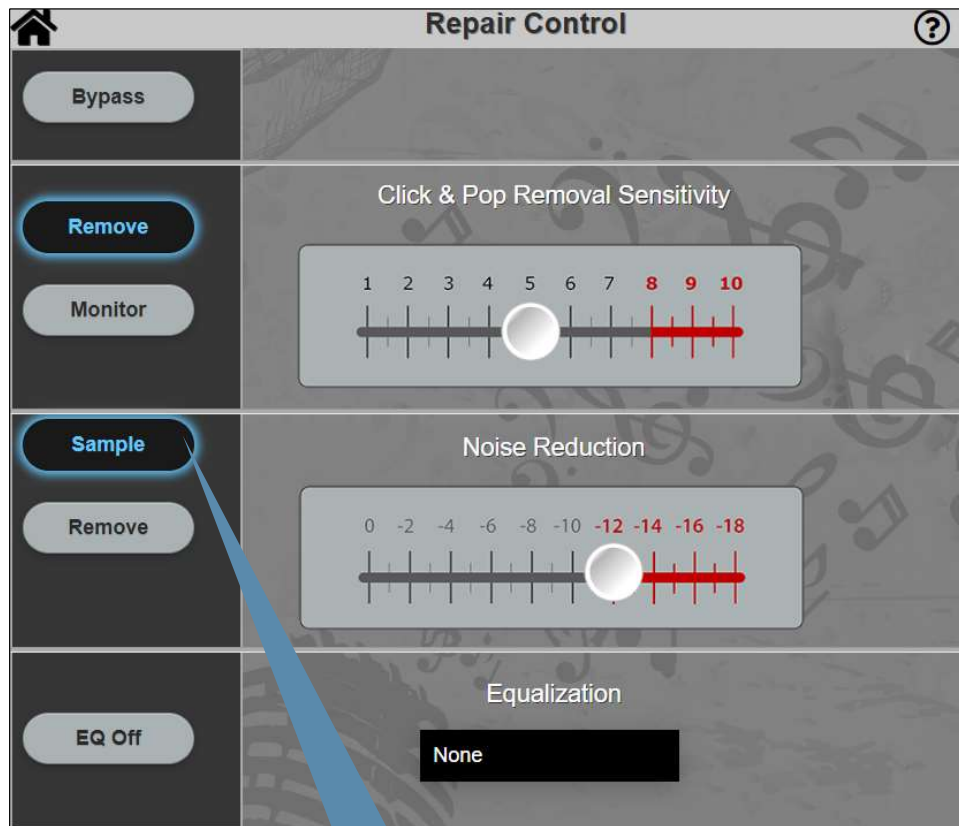


Setting of '4' or '5' is all you need for all but the most damaged LPs

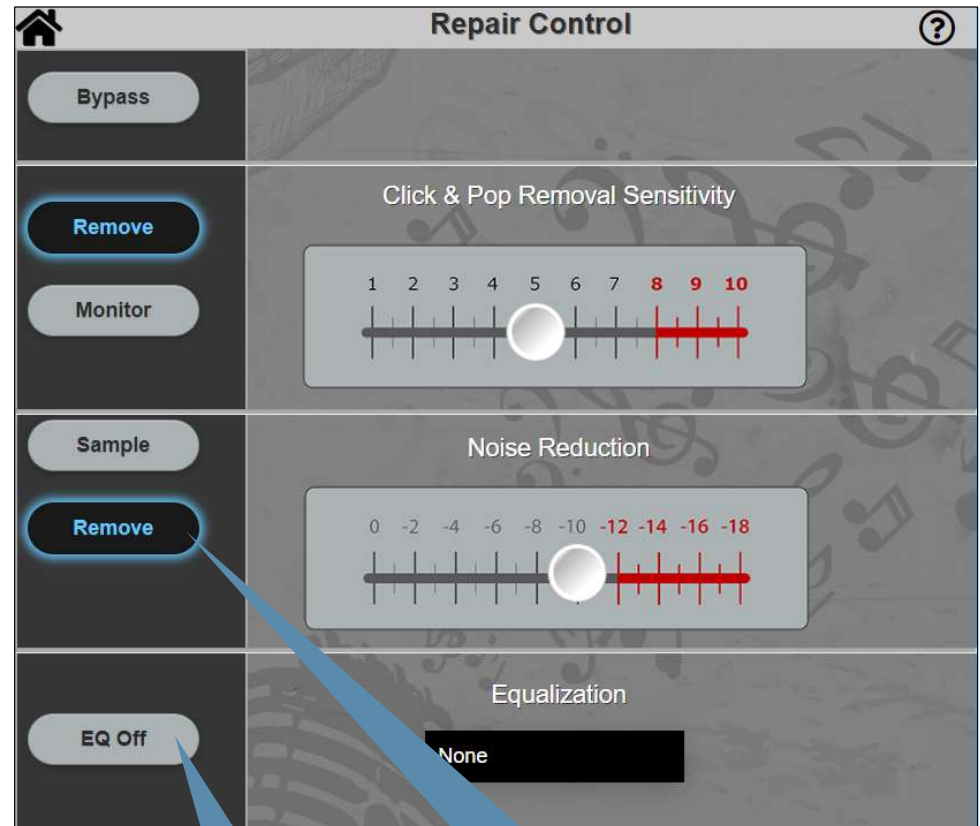
Monitor the clicks and pops that are being removed from the audio passing through the SugarCube.

4.3 Noise Reduction Functions

Controlling Noise Reduction



Click on '**Sample**' to initialize '**Noise Reduction**'. Drop the needle into the lead-in groove and the system takes a three second sample of vinyl noise from the lead in groove. When done, '**Sample**' turns off by itself.

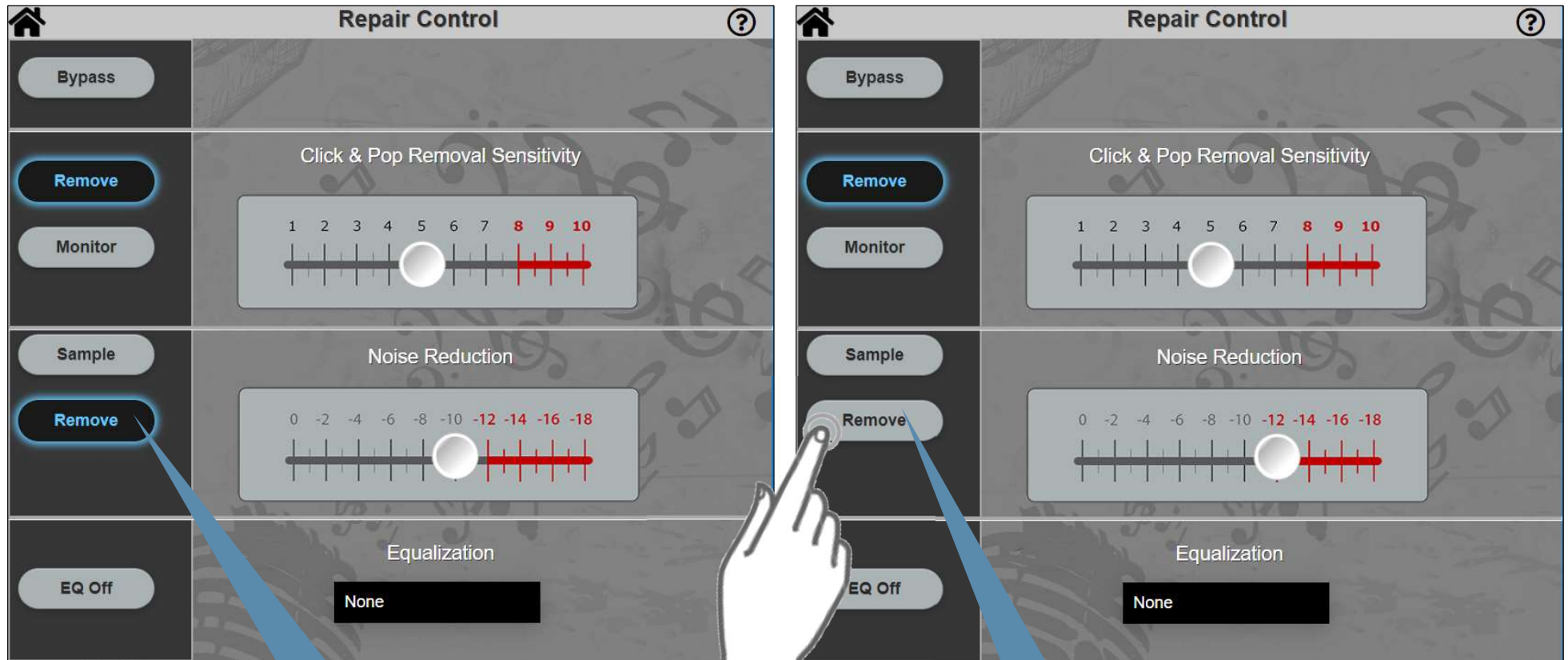


Note: **Record Equalization (EQ)** is off (in this example) and can be turned on at any time.

'**Noise Reduction**' engages/disengages noise reduction (note: setting the slider to '0' also disengages Noise Reduction.)

4.4 Noise Reduction Functions

Controlling Noise Reduction

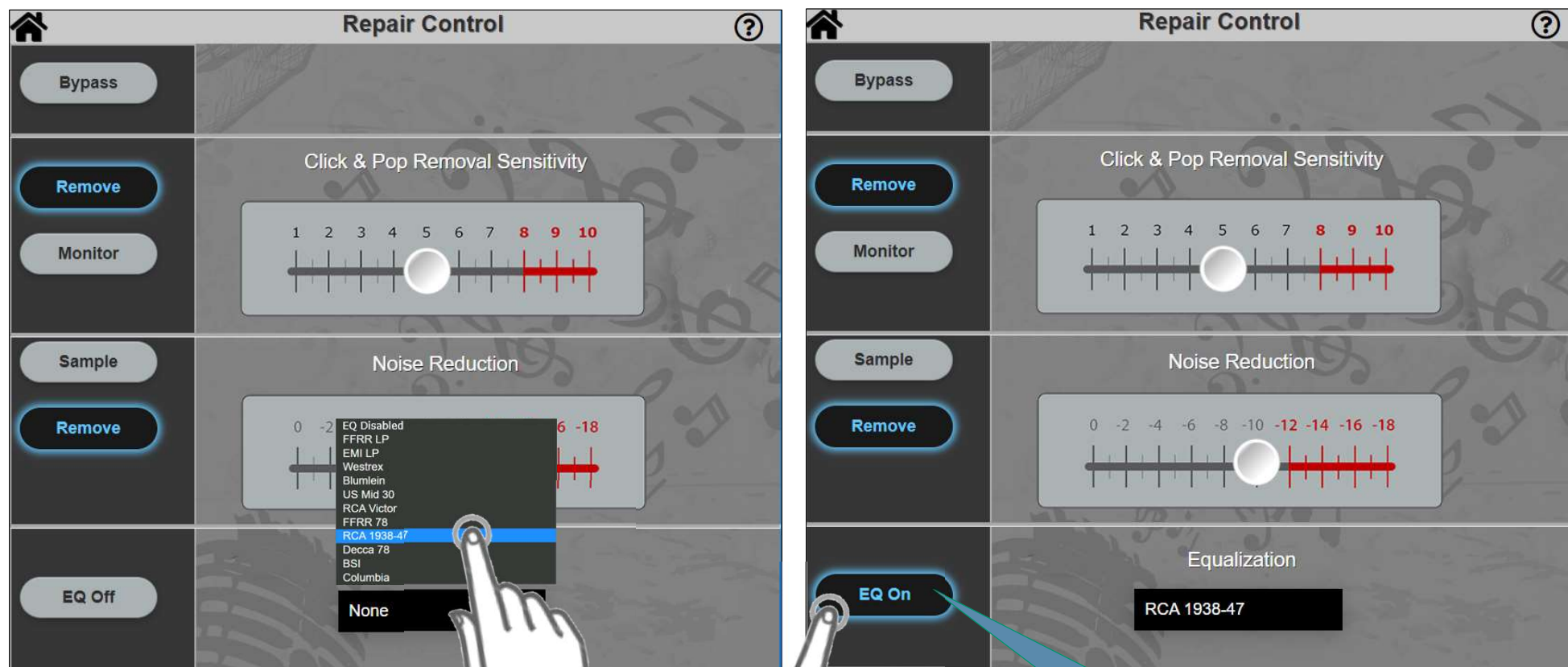


NR can be dis-engaged any time by clicking here or moving the slider to '0'.

Clicking on '**Remove**' disengages the NR process but you do not need to re-sample if the next vinyl LP is from the same time period and label.

4.5 Equalization (EQ) Compensation

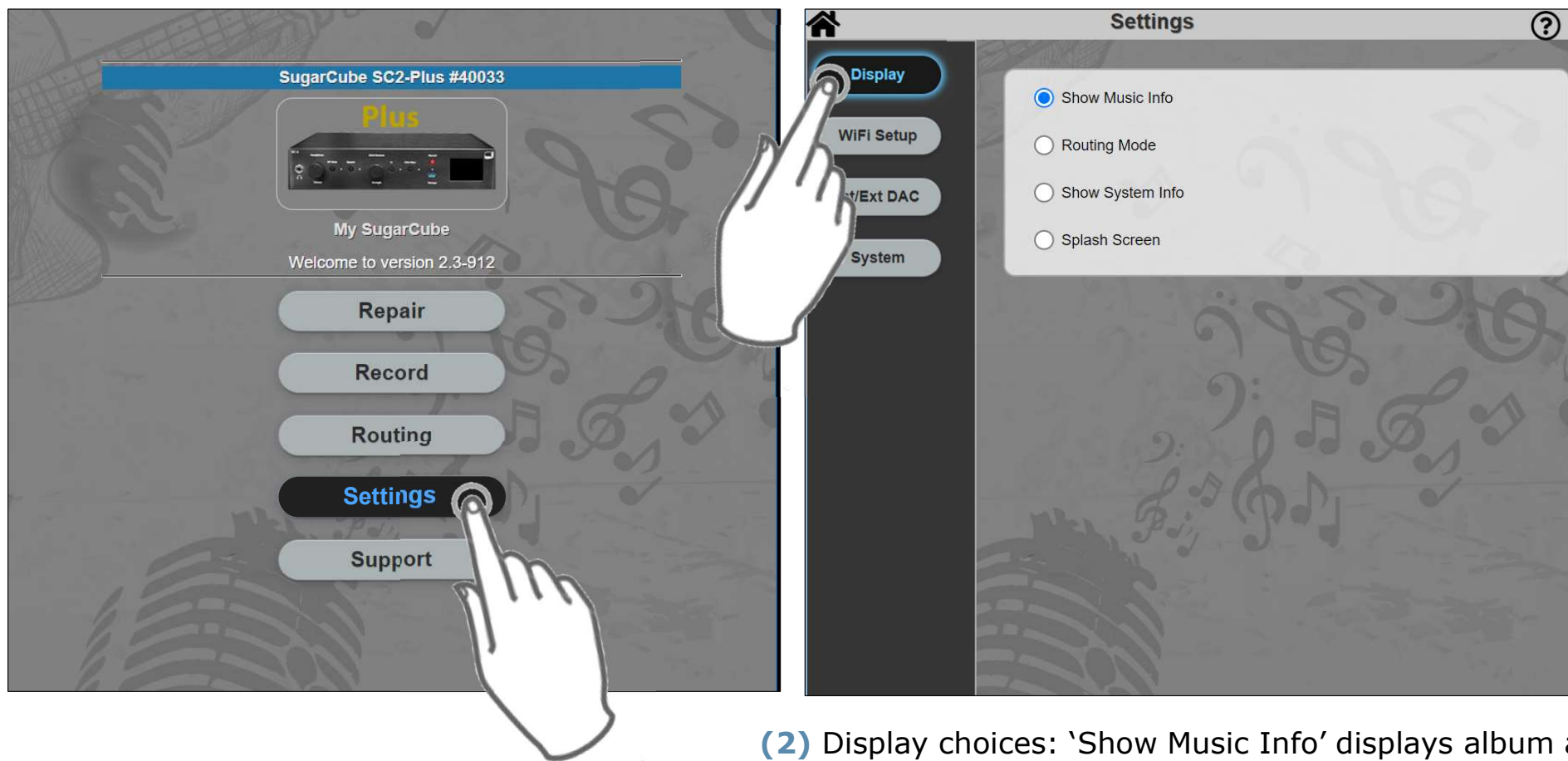
Controlling Playback EQ Curves



The desired EQ curve can be chosen from the pop-up menu.

Clicking on EQ will engage it at any time.

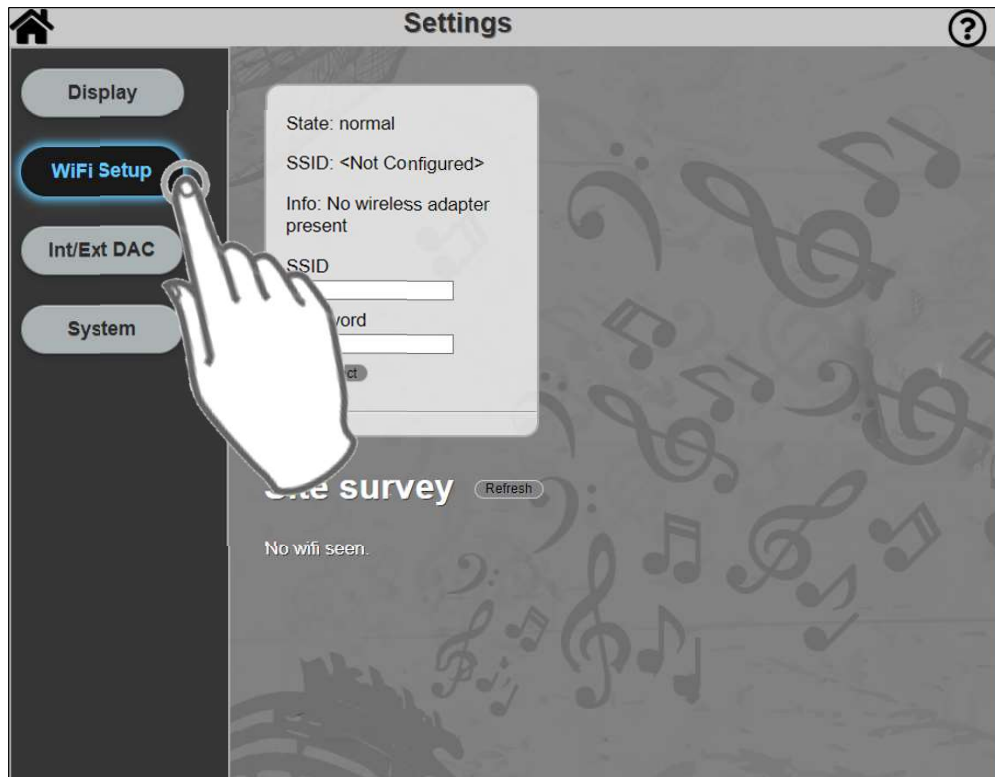
4.6 Settings and Display Functions



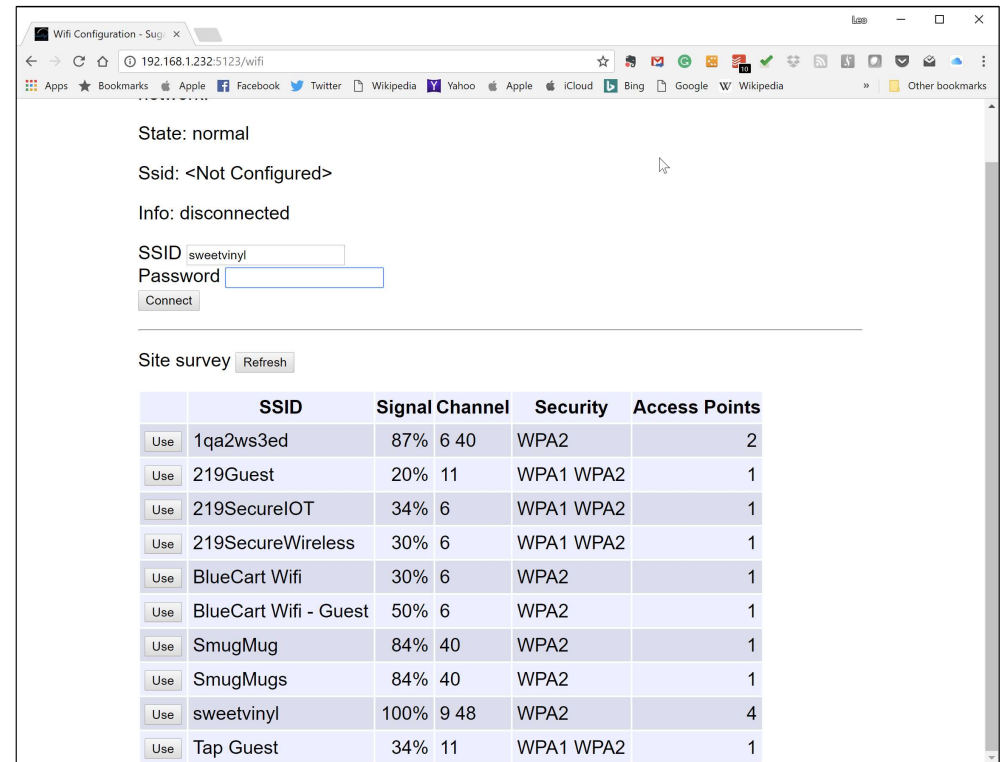
(1) Click on 'Settings'

(2) Display choices: 'Show Music Info' displays album art during recording; 'Routing Mode' display on the front panel display the current routing mode; 'System Info' shows **IP Address** and **Software Version**, can also be obtained from the front panel by pressings and holding the '**Bypass**' button for about **five seconds**.

4.7 Controlling the SugarCube via WiFi



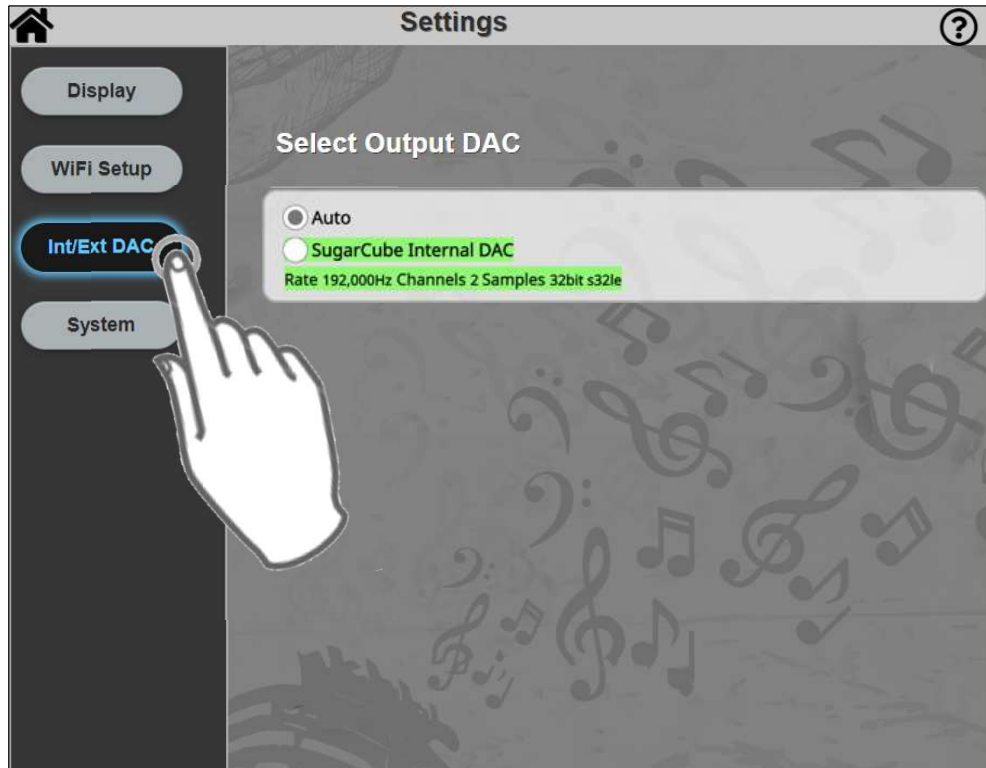
(1) Click on 'WiFi Setup'



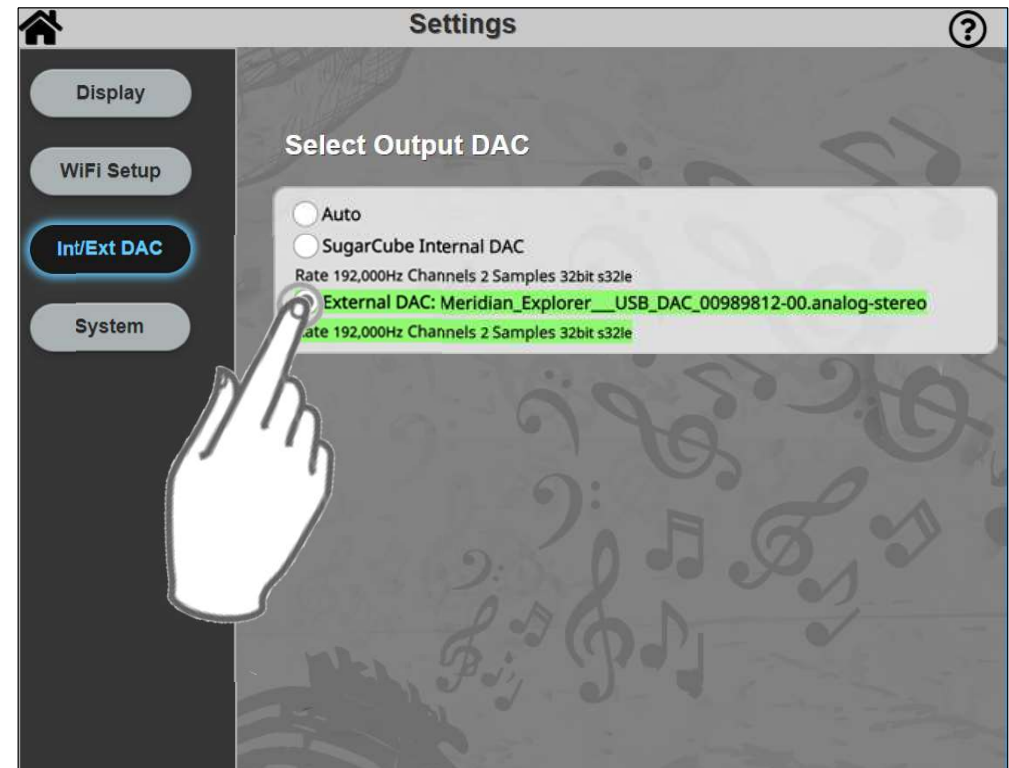
(2) When the Wifi dongle is in the rear USB port available networks will appear and you can select the SSID and enter your password.

4.8 Choosing an **External DAC** Output

Selecting an external DAC plugged into the SugarCube

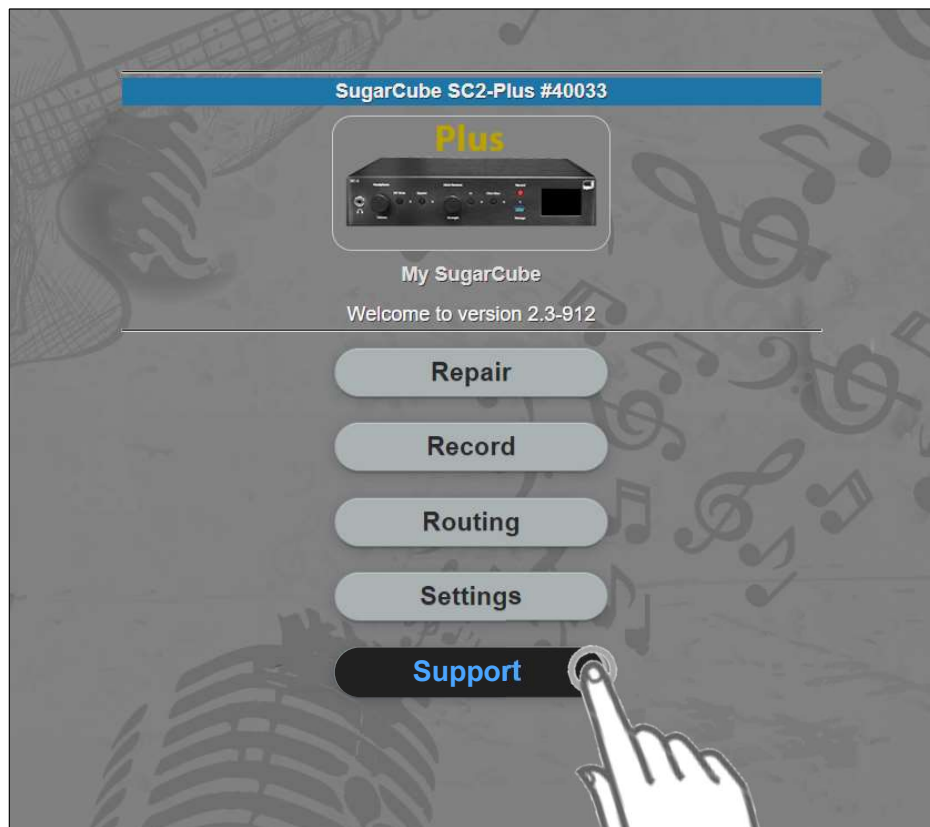


(1) Click on 'Int/Ext DAC.'

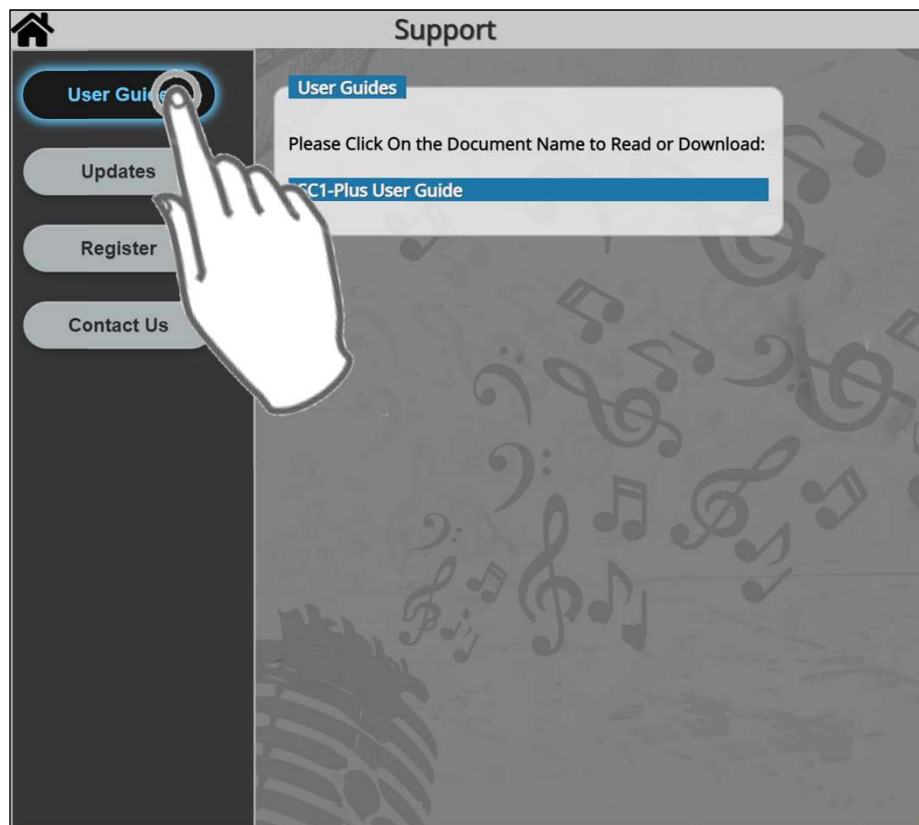


(2) When a USB DAC is inserted in one of the rear USB ports, it will appear here and can be selected instead of the internal DAC.

4.9 Support Menu & User Guide

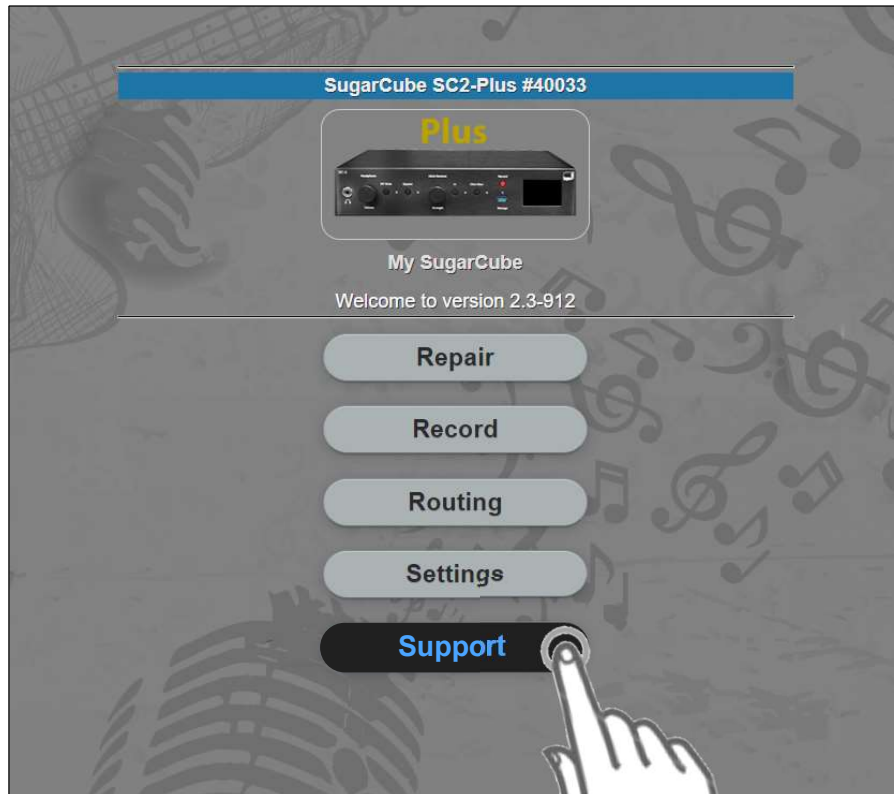


(1) Click on 'Support'

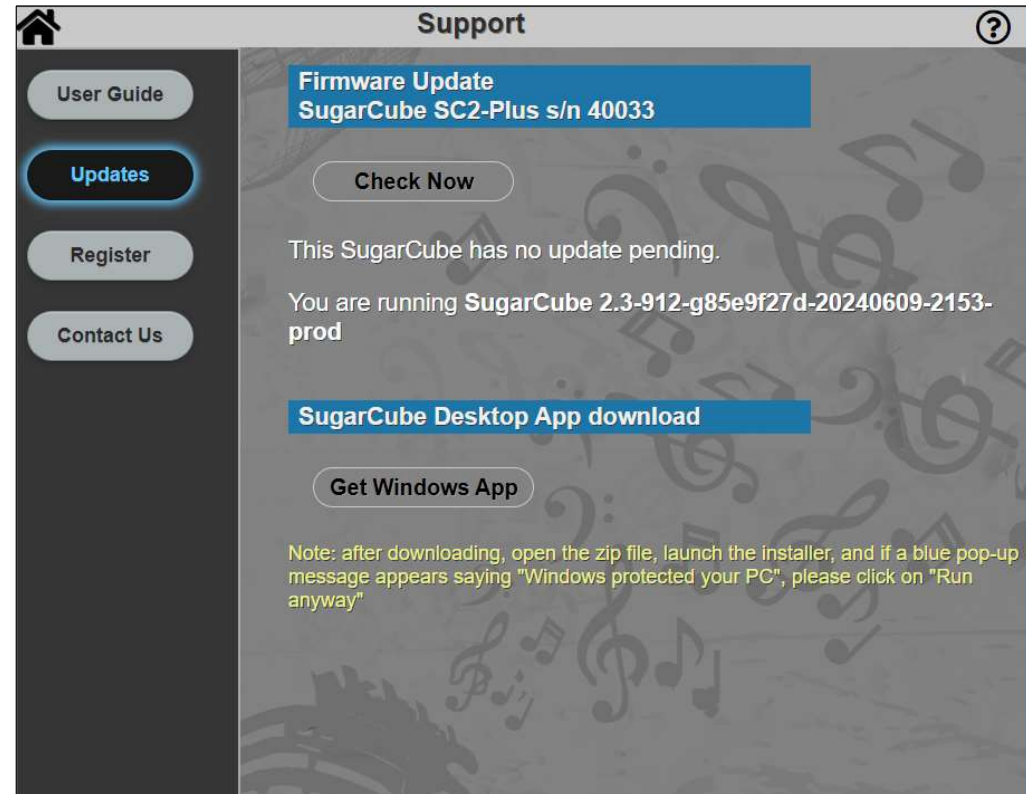


(2) The current User Guide for your model is available for download and/or viewing in the browser here.

4.10 Checking for Updates

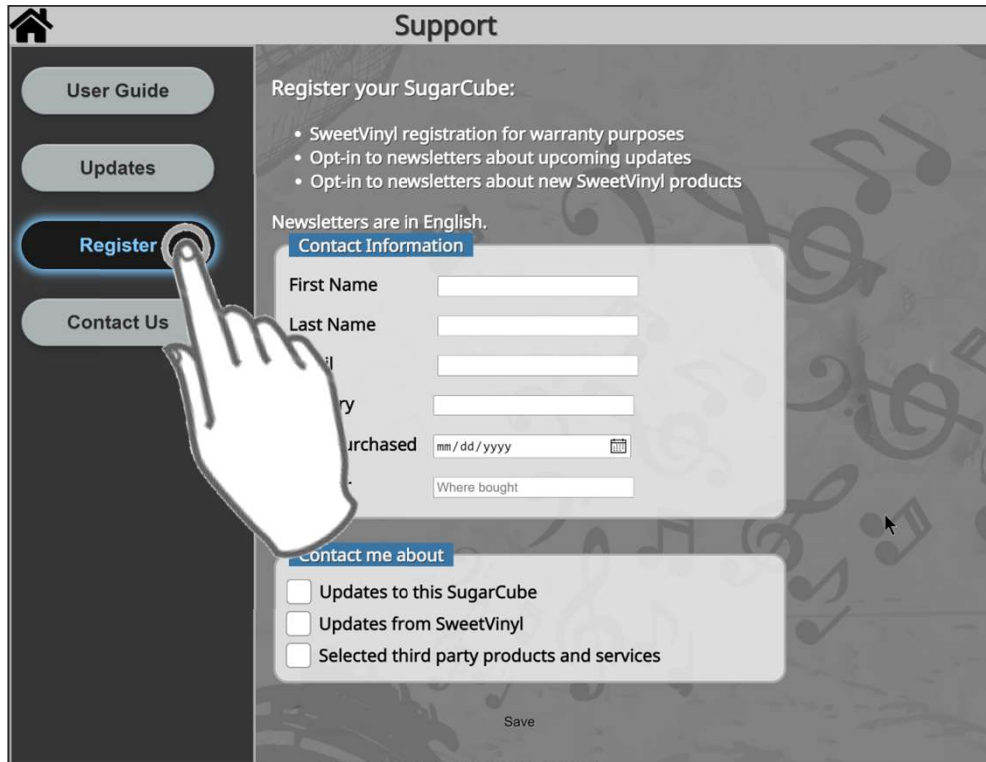


(1) Click on 'Support.'



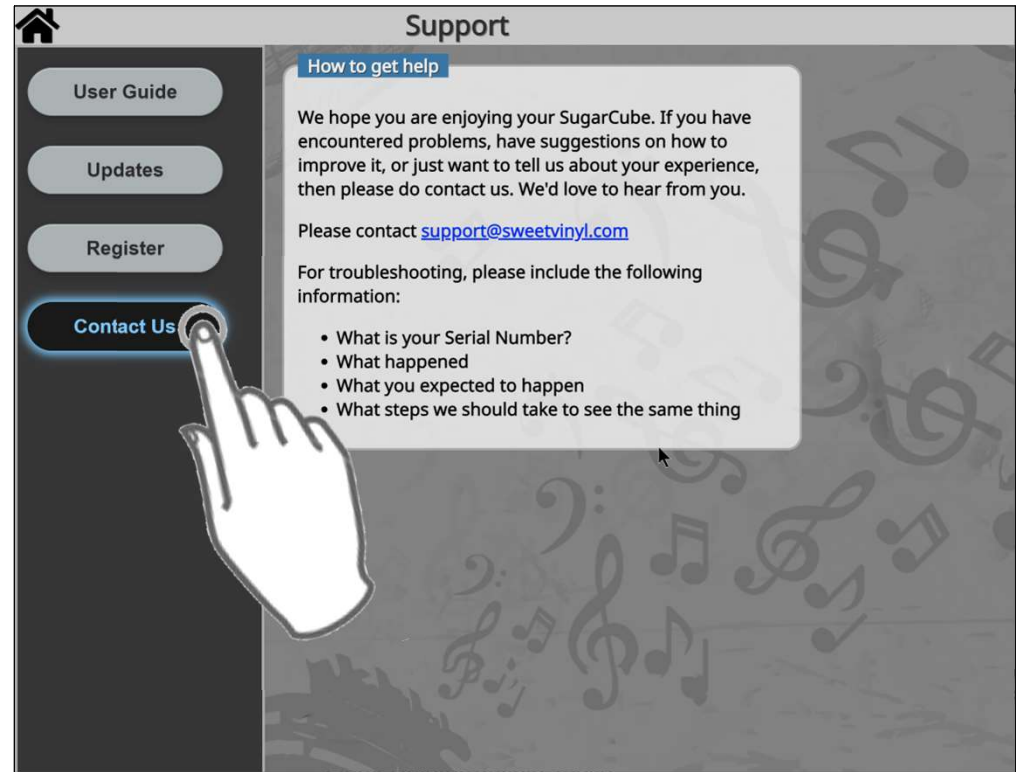
(2) Current SugarCube software version is displayed and updates can be checked. If updates are waiting can be updated here. Also shows last update and if an update is coming.

4.11 Register and Email Info



The screenshot shows a web interface titled 'Support'. On the left, there is a sidebar with buttons: 'User Guide', 'Updates', 'Register', and 'Contact Us'. The 'Register' button is highlighted with a hand cursor. The main content area is titled 'Register your SugarCube:' and contains a list of bullet points: 'SweetVinyl registration for warranty purposes', 'Opt-in to newsletters about upcoming updates', and 'Opt-in to newsletters about new SweetVinyl products'. Below this, it says 'Newsletters are in English.' and 'Contact Information'. There are input fields for 'First Name', 'Last Name', 'Email', 'Phone', 'Address', 'City', 'State', 'Zip', 'Purchased' (with a date picker), and 'Where bought'. At the bottom, there is a 'Contact me about' section with checkboxes for 'Updates to this SugarCube', 'Updates from SweetVinyl', and 'Selected third party products and services'. A 'Save' button is at the bottom right.

(1) Click on 'Register' and you can enter your information.



The screenshot shows the same 'Support' page, but now the 'Contact Us' button in the sidebar is highlighted with a hand cursor. The main content area is titled 'How to get help' and contains a paragraph: 'We hope you are enjoying your SugarCube. If you have encountered problems, have suggestions on how to improve it, or just want to tell us about your experience, then please do contact us. We'd love to hear from you.' Below this, it says 'Please contact support@sweetvinyl.com'. Then, it says 'For troubleshooting, please include the following information:' followed by a list of bullet points: 'What is your Serial Number?', 'What happened', 'What you expected to happen', and 'What steps we should take to see the same thing'.

(2) 'Contact Us' will take you to a screen where you can follow instructions to send an email to support@sweetvinyl.com.

Section 5

Connecting a Computer over USB

MacOS, Windows,
or Music Server



5.1 Connecting a Computer over USB

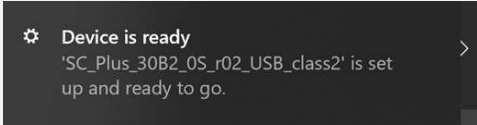


The SC2-Plus can be connected to a computer via the **USB-B (Host)** port. Plug one end into the USB port on the computer and the other to the **USB-B** port on the SugarCube. When connected, the computer will be able to control the **ADC** and **DAC** functions on the SugarCube. The computer can record the output of the **ADC** as well as play back through the **DAC** with any digital audio software.

5.1.1 Configuring Windows to control the SugarCube

1

(1) After connecting your Windows PC, the SugarCube should configure itself



Device is ready
'SC_Plus_30B2_0S_r02_USB_class2' is set up and ready to go.

2

(2) Drivers are built in and you should be able to see and configure the SugarCube as a device under Sounds by right clicking on the speaker Icon on the task bar.

Open Sound settings

Open Volume mixer

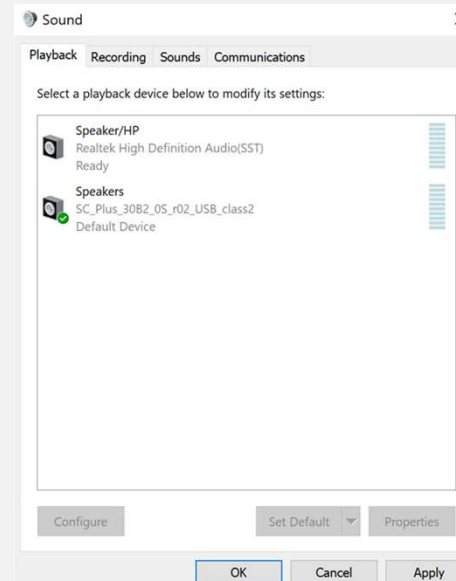
Spatial sound (Off)

Sounds

Troubleshoot sound problems

3

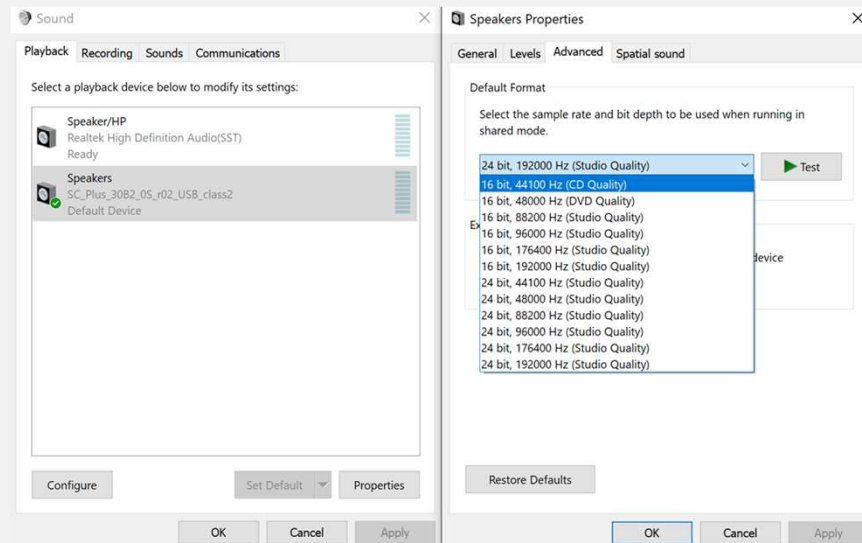
(3) A Window like this will open:



5.1.2 Configuring Windows to control the SugarCube

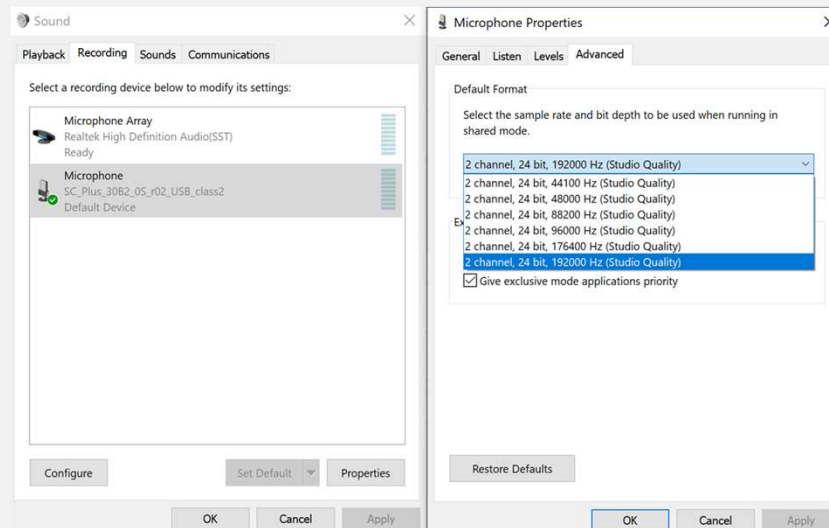
4

(4) As a Playback device you will be able to configure the SugarCube DAC as a Speaker – by selecting the SC Plus you can then select Properties and the Advanced tab and select the sample rate and bit depth. You will want to set this to 24 bit, 192K if you want the SugarCube functions to work.



5

(5) As a Recording device the SugarCube ADC will show up as a Microphone by selecting the SC Plus you can then select Properties and the Advanced tab and select the sample rate and bit depth. You will want to set this to 24 bit, 192K if you want the SugarCube functions to work.



5.1.3 Configuring MacOS to control the SugarCube

1

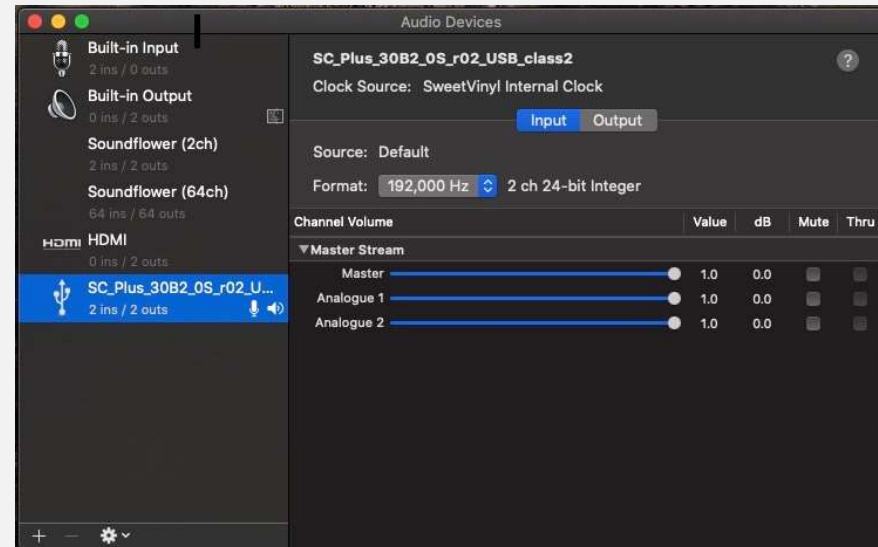
(1) After connecting your MacOS computer, the SugarCube should configure itself

2

(2) Drivers are built in and you should be able to see and configure the SugarCube as a device under "Audio Midi Setup" by searching for this or by going to the "Utilities" folder in "Applications"

3

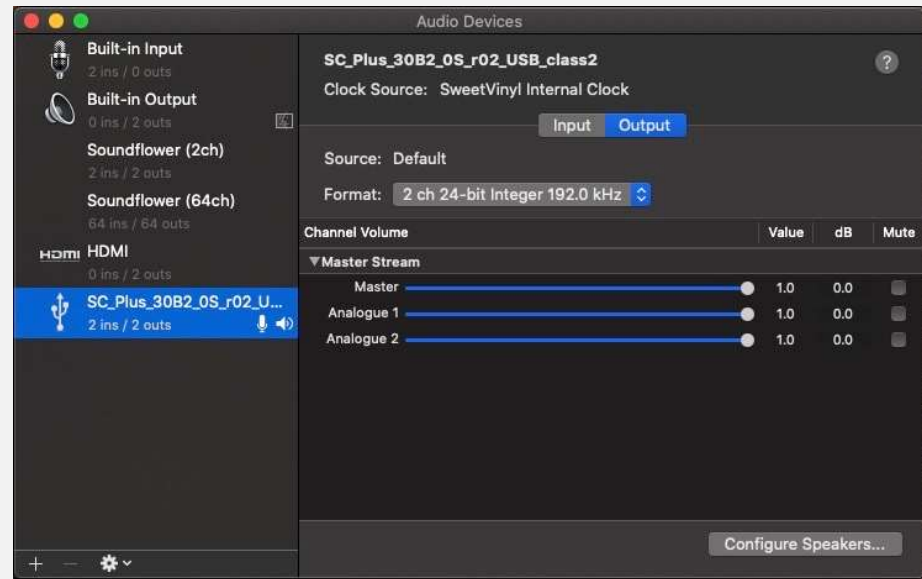
(3) A Window like this will open and you will want to further select the "Audio Setup" section:



5.1.4 Configuring MacOS to control the SugarCube

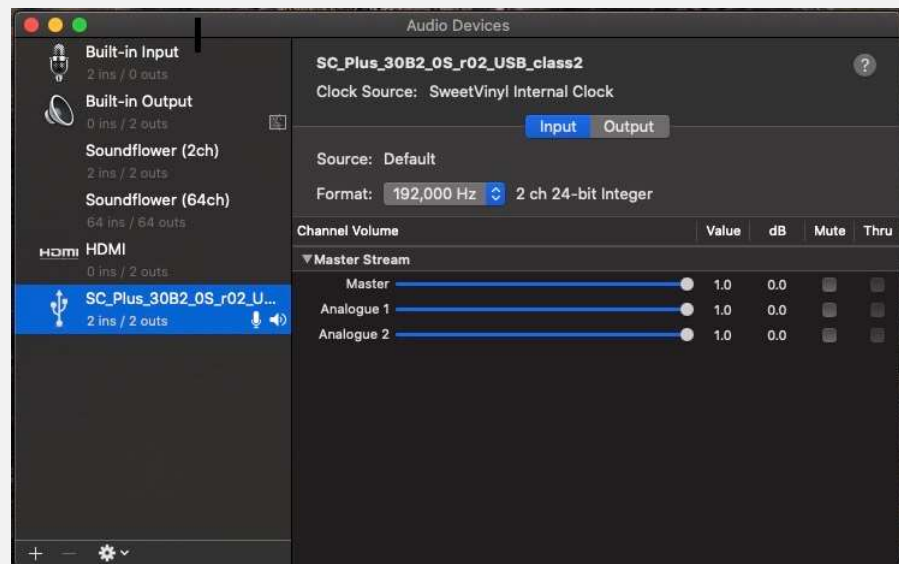
4

(4) As an Output device you will be able to configure the SugarCube DAC– by selecting the SC Plus you can the sample rate and bit depth. You will want to set this to 24 bit, 192K if you want the SugarCube functions to work.

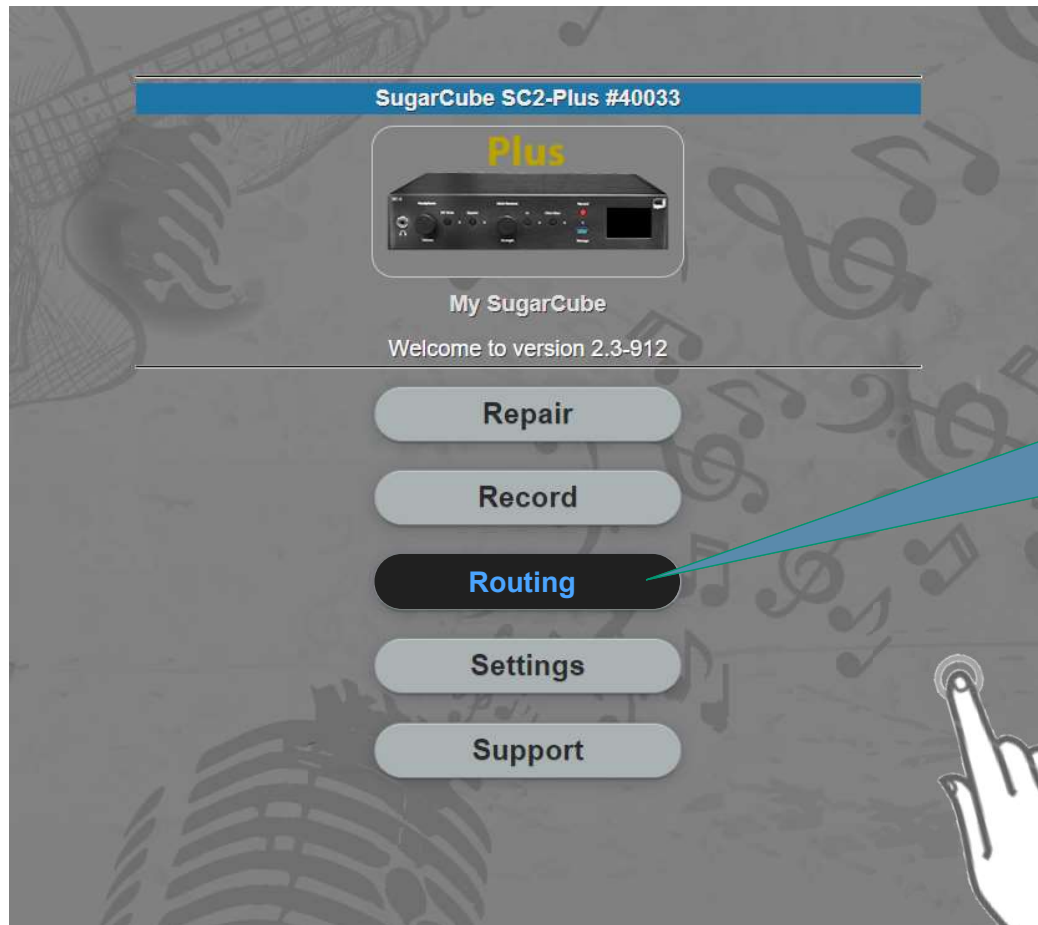


5

(5) As an Input device the SugarCube ADC will appear and by selecting the SC Plus you can select the sample rate and bit depth. You will want to set this to 24 bit, 192K if you want the SugarCube functions to work.

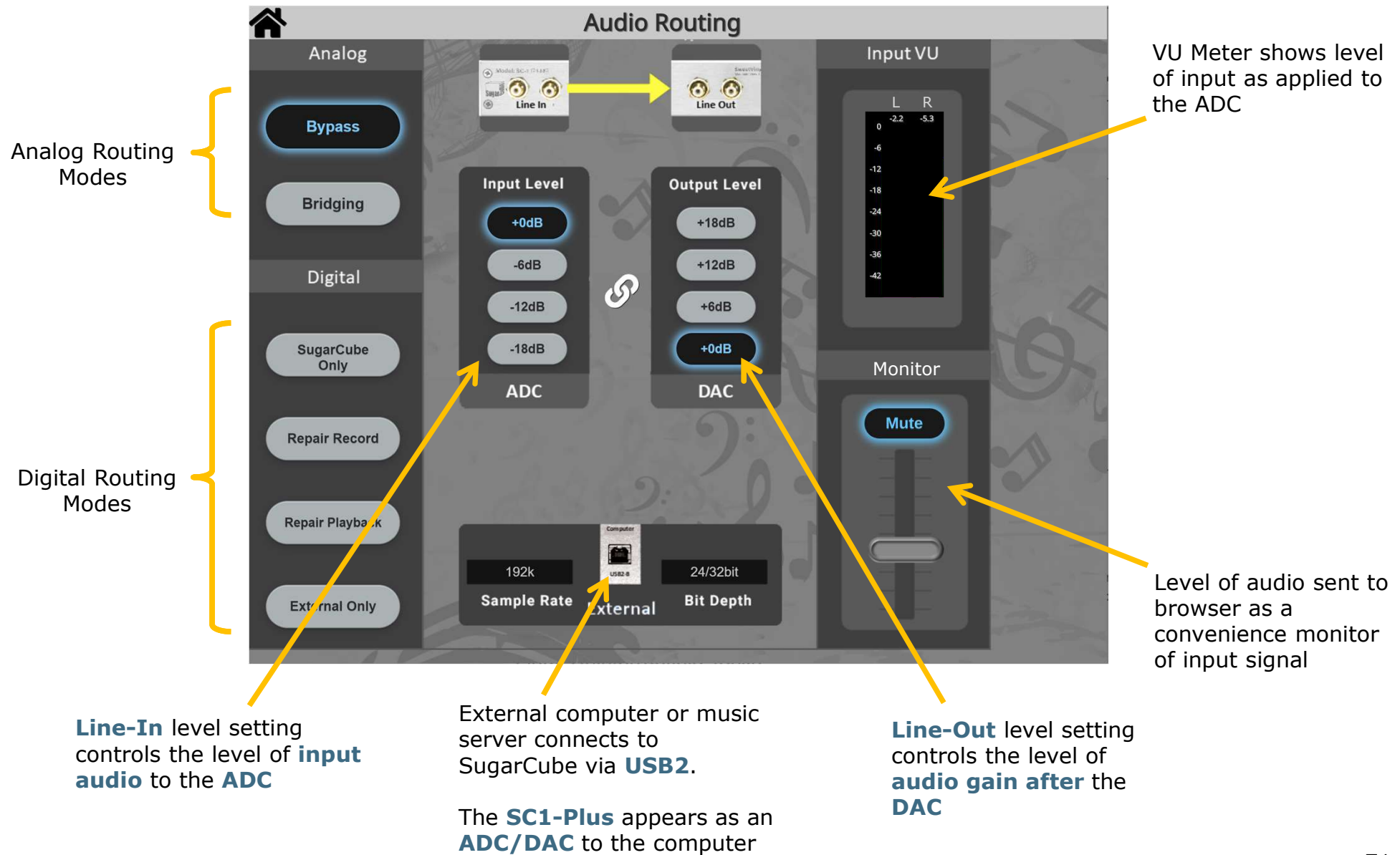


5.2 Audio Routing Menu

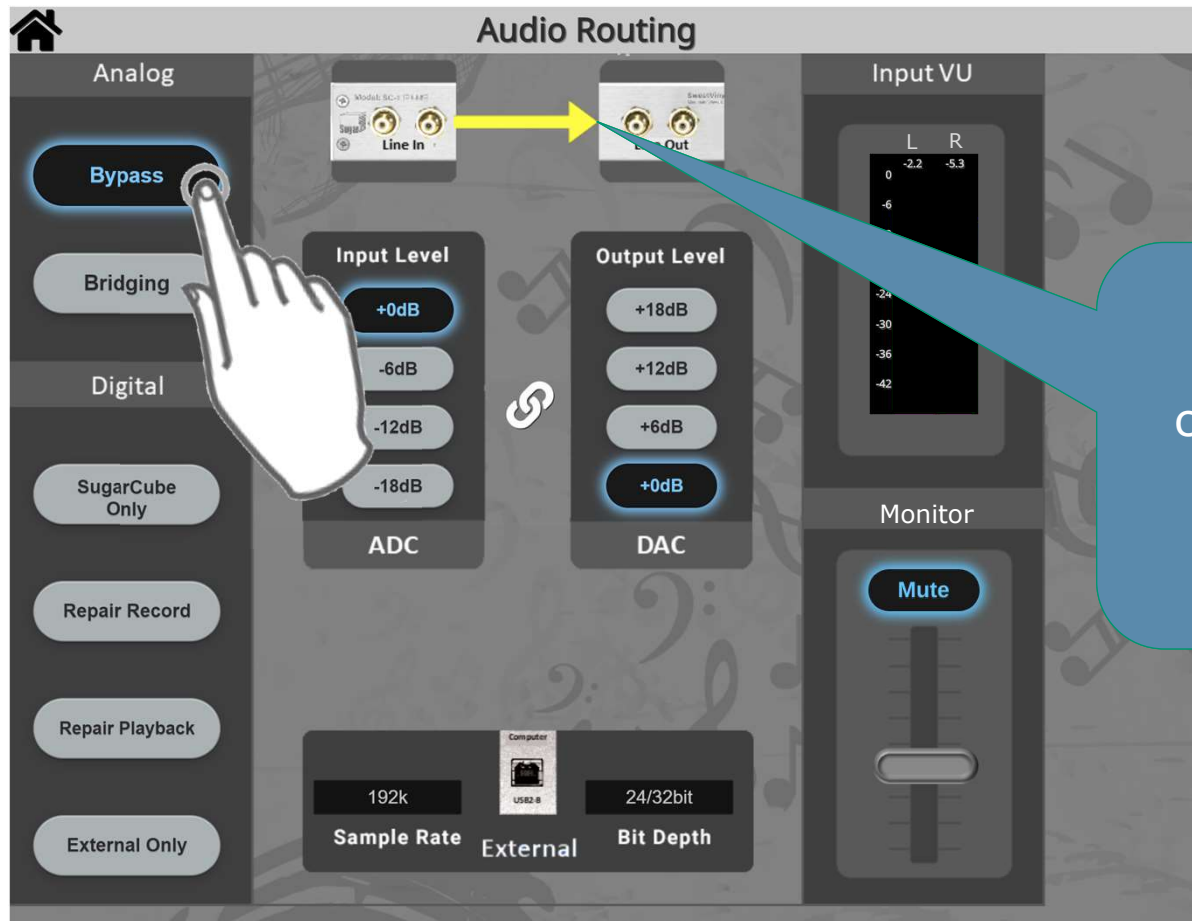


To efficiently work with the SC1-Plus and a computer connected over USB you will want to open the **Routing Menu** page. The next pages show how the SugarCube, and an external computer, work together to provide the enhanced functionality.

5.3 Audio Routing Menu — Control Layout



5.4 Audio Routing Menu — Bypass Mode



When **Bypass** is selected, **Line-In** connects directly to **Line-Out** thru an audiophile-grade mechanical relay.



In **Bypass** mode, **Line In** connects to **Line Out** via a mechanical relay; this can be selected here or on the front panel. When in **Bypass** no signal is processed by the SugarCube nor by a connected computer.

5.5 Logical Flow — **Bypass** Mode

Line-In connects directly to Line-Out.



Bypass

Bridging

Mode 1

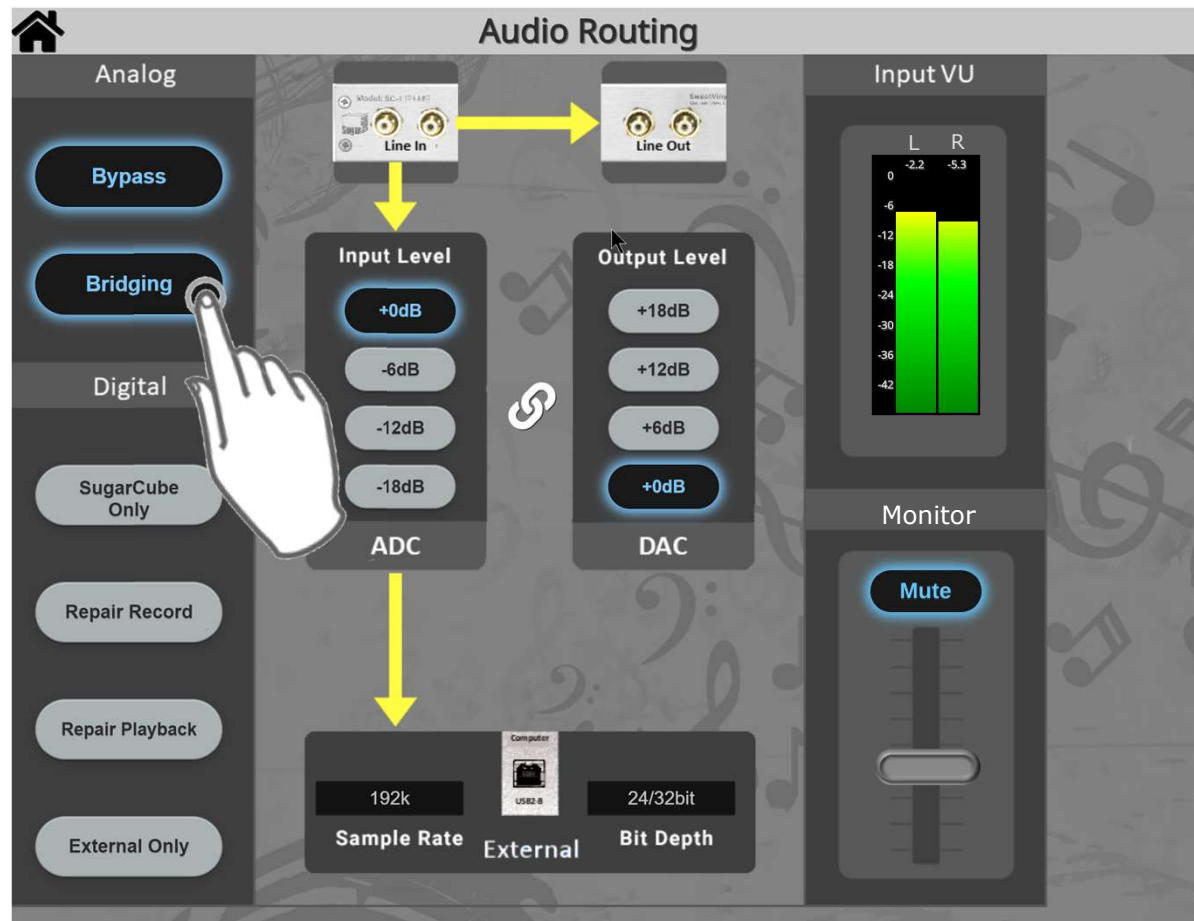
Mode 2

Mode 3

Mode 4

In Bypass mode **Line-In** is connected to **Line-Out** via a mechanical relay, this can be selected on the menu or on the front panel. When in **Bypass**, no signal is processed by the SugarCube or a connected computer.

5.6 Audio Routing Menu — Bridging Mode



There is no audible affect as **Line-In** passes directly to **Line-Out**.

Bridging mode passes **Line-In** to the Analog-to-Digital Converter (**ADC**) which passes its digital output signal to the attached computer over **USB**.

The **ADC** operates at 192k/24bits by **default** but the external computer can change to any standard sample rate and bit depth.

Bypass

Bridging

Mode 1

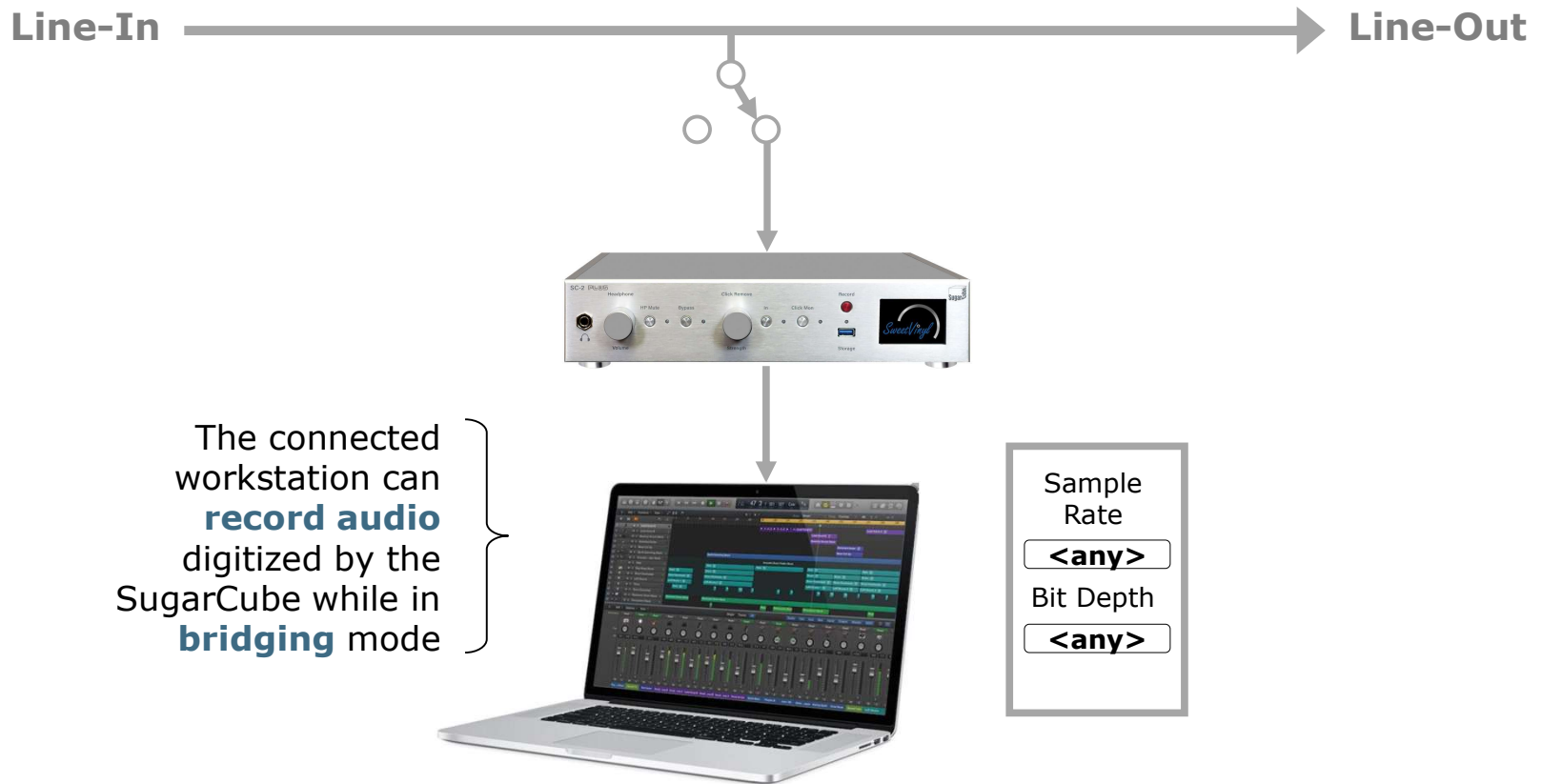
Mode 2

Mode 3

Mode 4

5.7 Logical Flow — Analog Bridging Mode

Line-In connects directly to Line-Out and SugarCube bridges the signal (with high impedance.)



Bypass

Bridging

Mode 1

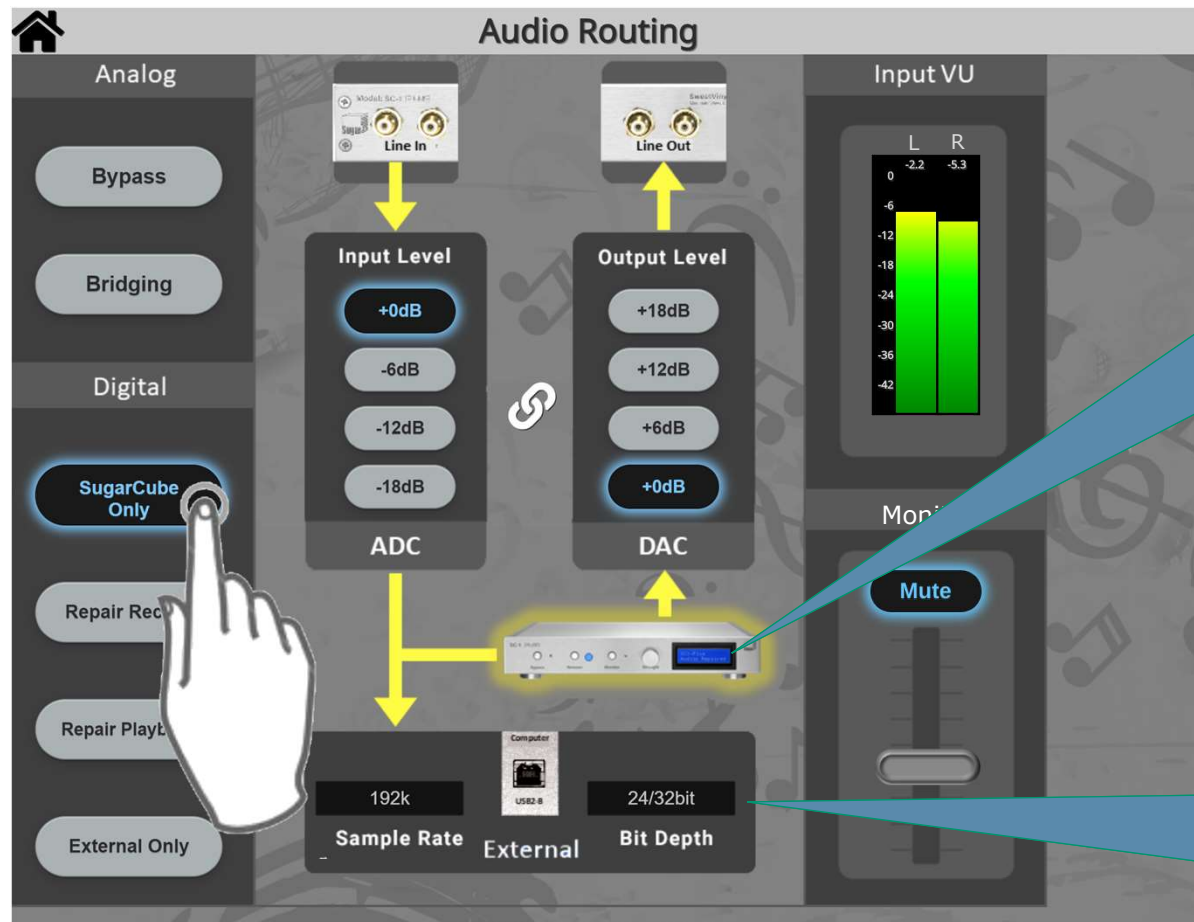
Mode 2

Mode 3

Mode 4

In **Bridging Mode**, **Line-In** is connected to **Line-Out** as in Bypass mode, in addition the signal is passed to the attached computer via the SugarCube's **ADC**.

5.8 Audio Routing Menu — SugarCube Only (Mode 1)



Audio Passes from **Line-In** through the **ADC** then into the **SugarCube** then into the **DAC** to **Line-Out**

Raw digital audio from the **ADC** output is also sent over **USB** to the attached computer at 192K/24bit

Bypass

Bridging

Mode 1

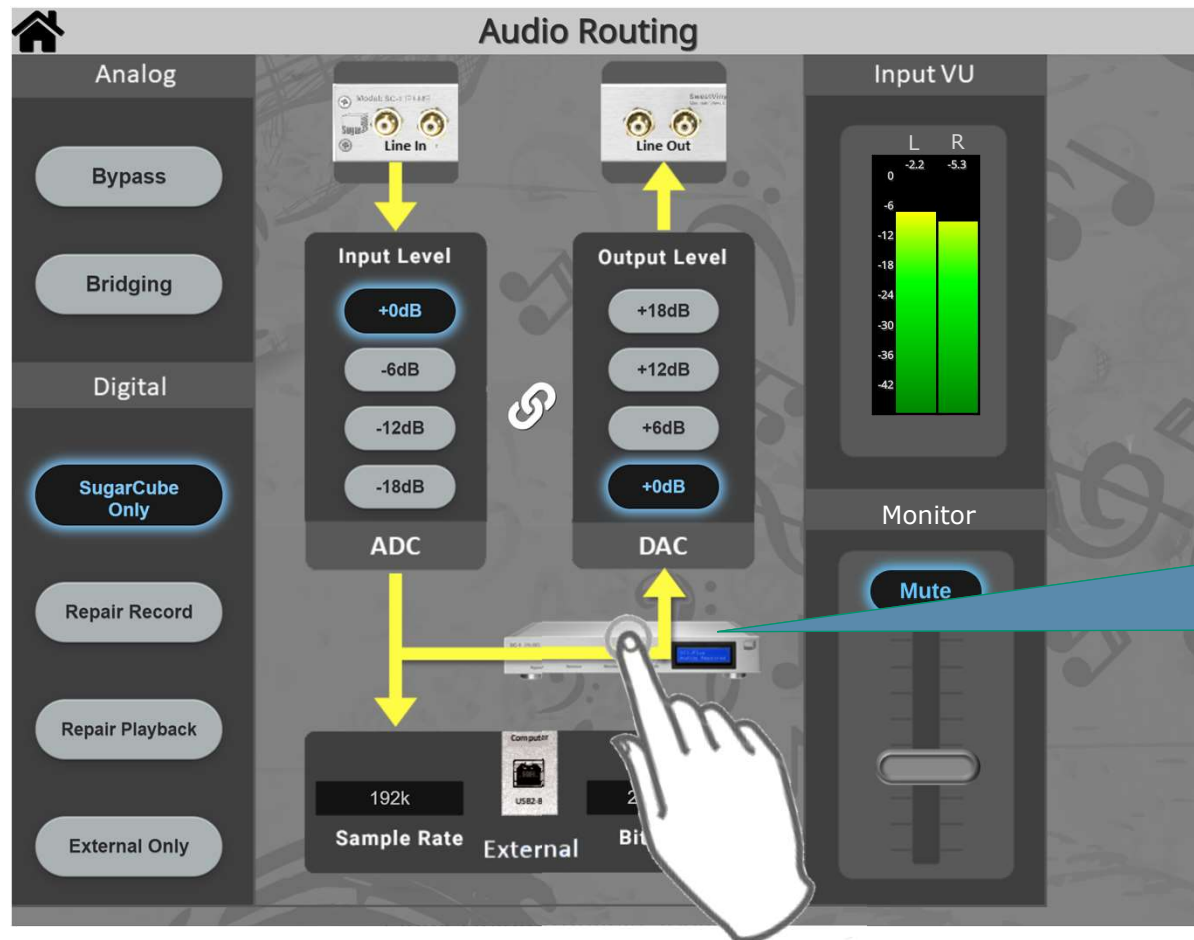
Mode 2

Mode 3

Mode 4

In **SugarCube-Only** mode, audio is processed by the SugarCube as standard on all Sugar Cubes. In addition, the **Un-Repaired Digital** signal is passed to the attached computer via the **ADC**.

5.9 Audio Routing Menu — SugarCube Only (Mode 1)



Click on the SugarCube **icon** to engage/dis-engage repair functions

Bypass

Bridging

Mode 1

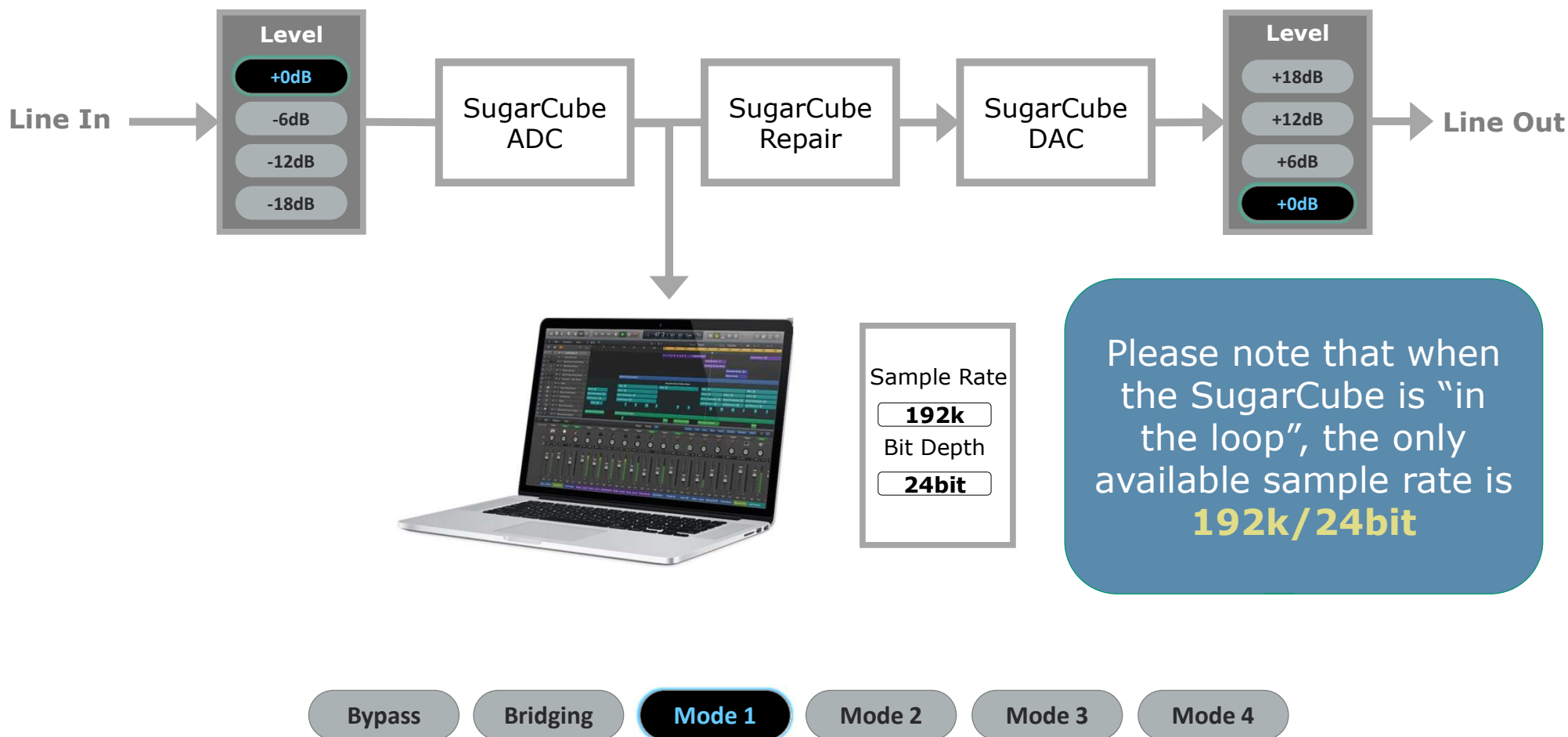
Mode 2

Mode 3

Mode 4

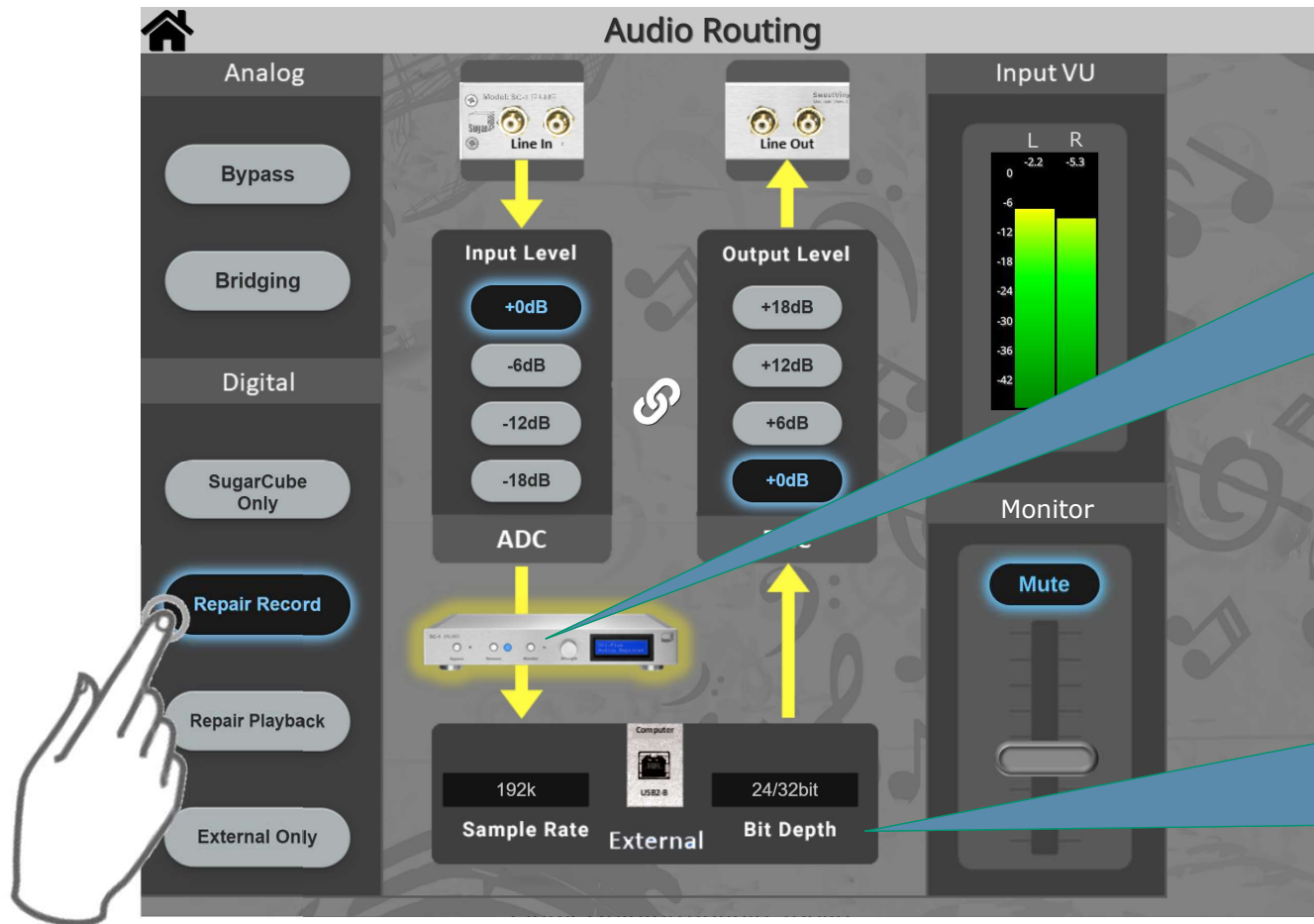
5.10 Logical Flow — SugarCube Only (Mode 1)

Normal SugarCube operations but with digital signal from ADC tapped and send out to workstation.



In **SugarCube-Only** mode, **Line-In** is processed by the SugarCube and audio is repaired and sent to **Line Out** as standard on all SugarCubes. In addition the repaired signal is passed to the attached computer via the **ADC**.

5.11 Audio Routing — Repair Record (Mode 2)



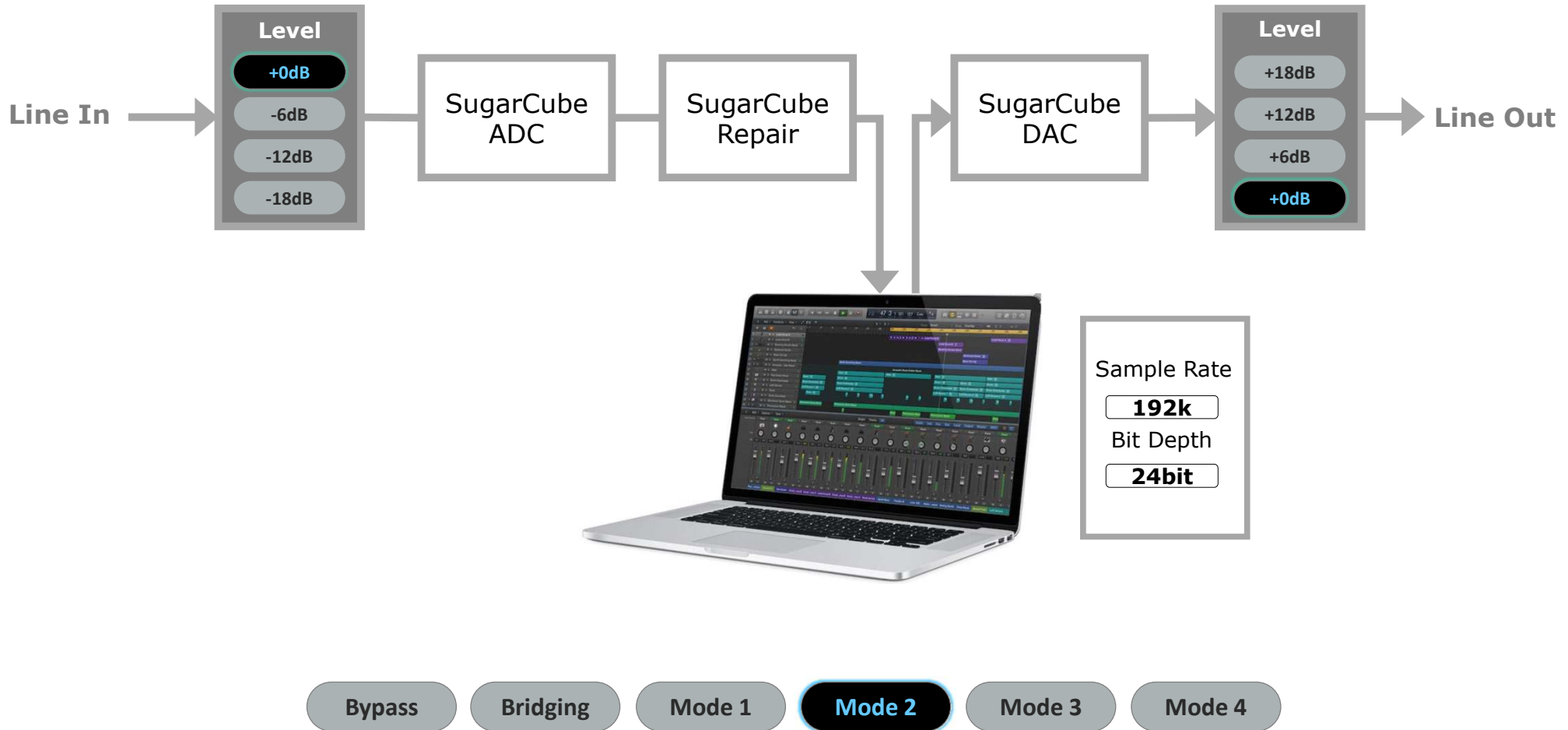
'**Repair Record**' inserts the SugarCube repair processing **after** the **ADC** to repair digital audio before outputting to the external device.

If the connected computer operates at **192k/24bit**, the system will be able to utilize SugarCube repair functions in line with the digital audio.

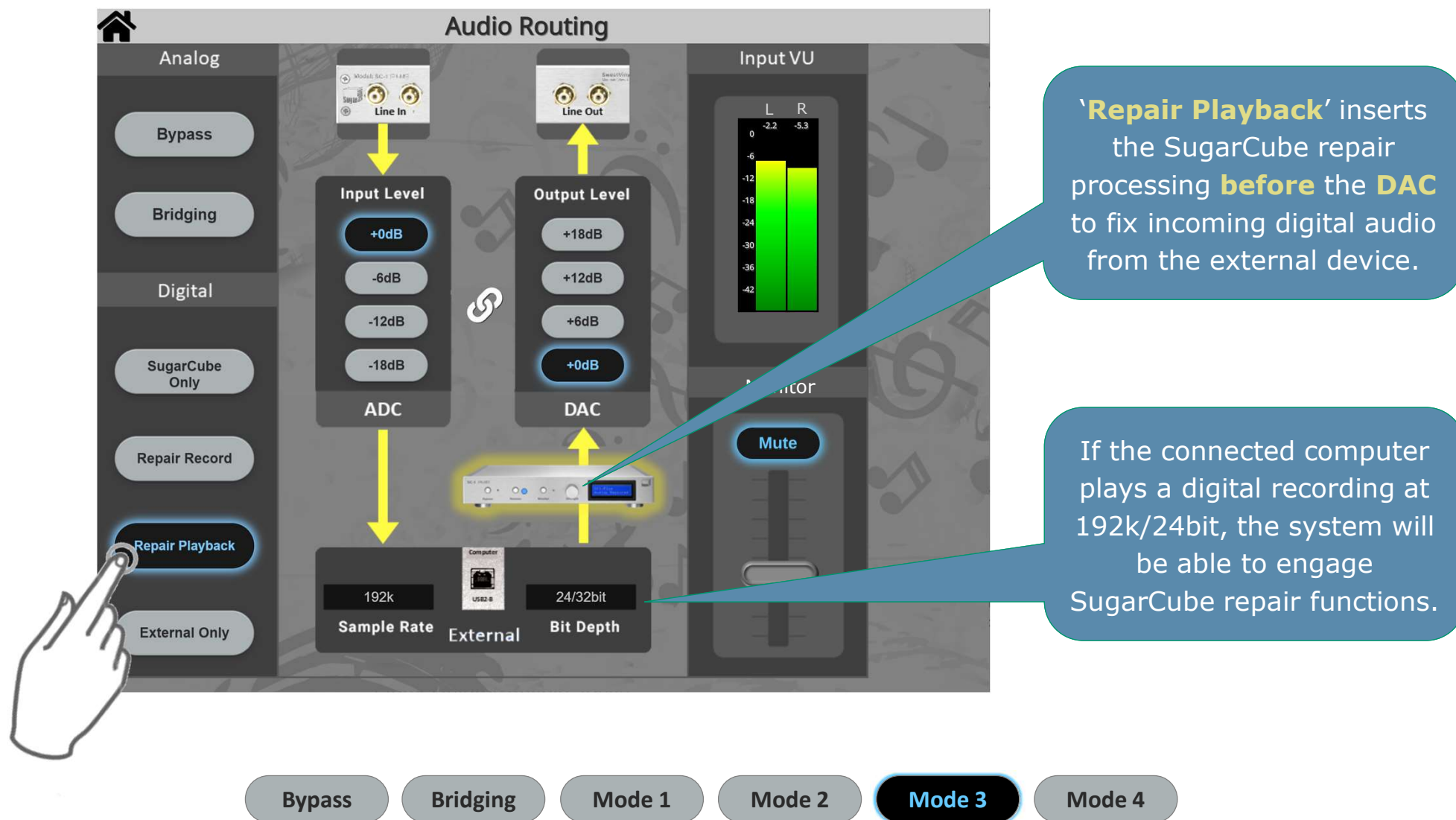
Note: Please **press again** the button of **any mode** of operation (even though it is already selected) if the external device **loses lock** with the SugarCube (you will certainly know due to obvious digital noise.) This resets the internal timing and restores the audio.

5.12 Logical Flow — Repair Record (Mode 2)

**SugarCube repairs audio before sending out to workstation.
Workstation directly drives the DAC.**



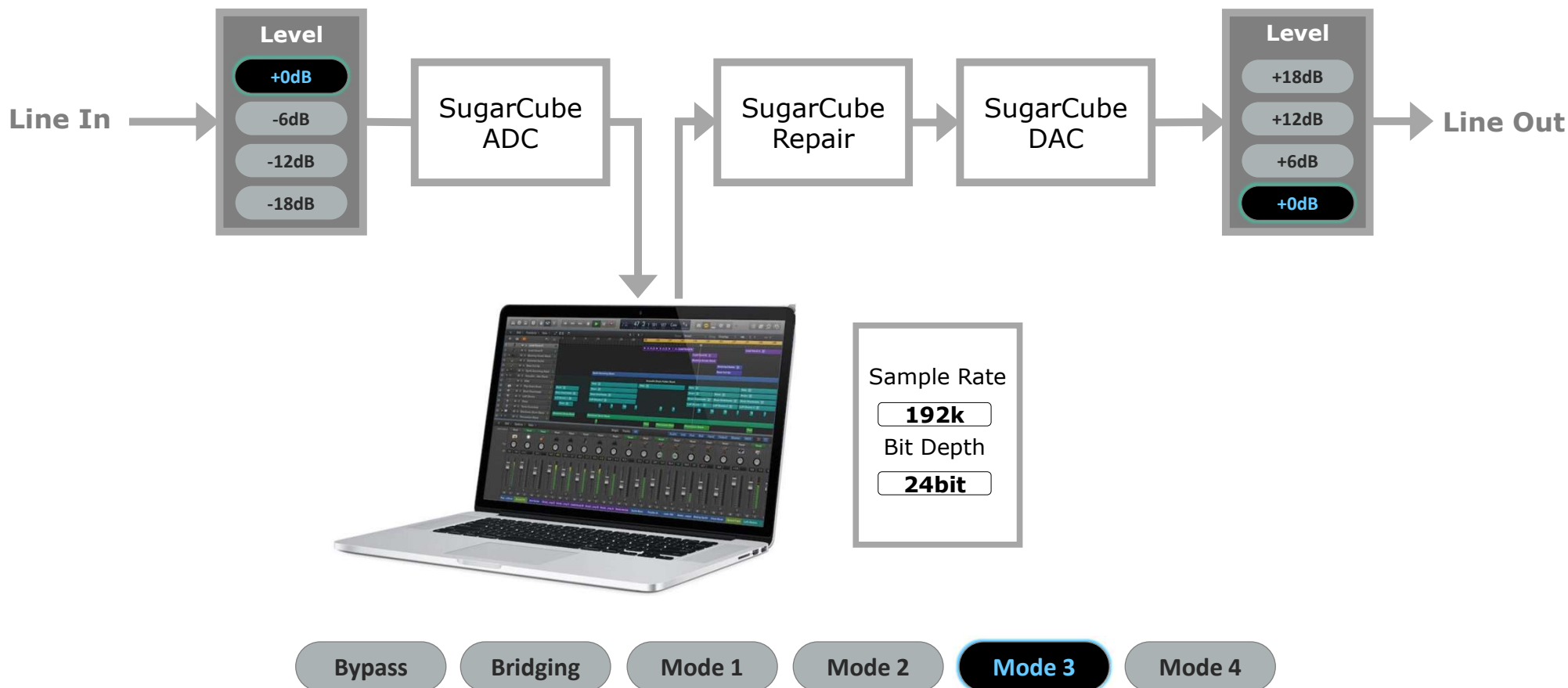
5.13 Audio Routing — Repair Playback (Mode 3)



The **Repair Playback** mode is particularly useful allowing you to play any digital file from your library to remove clicks & pops and/or remove tape hiss from previous recordings from any source from the past. 81

5.14 Logical Flow — Repair Playback (Mode 3)

Digitized audio goes directly to workstation. Workstation output feeds SugarCube for repair before DAC.



Repair Playback is our **recommended mode** for operation. All of your files will be **recorded** “in the raw” with no SugarCube processing applied. Then, on **playback**, you can adjust click & pop removal as well as apply noise reduction (if needed) on a recording-by-recording basis or use no processing at all if desired.

5.15 Audio Routing Menu — External Only (Mode 4)

Audio Routing

Analog

- Bypass
- Bridging

Digital

- SugarCube Only
- Repair Record
- Repair Playback
- External Only**

Line In → **Input Level** (+0dB, -6dB, -12dB, -18dB) → **ADC** → **Sample Rate** (192k) → **External** → **Bit Depth** (24/32bit) → **DAC** → **Output Level** (+18dB, +12dB, +6dB, +0dB) → **Line Out**

Input VU

Monitor

Mute

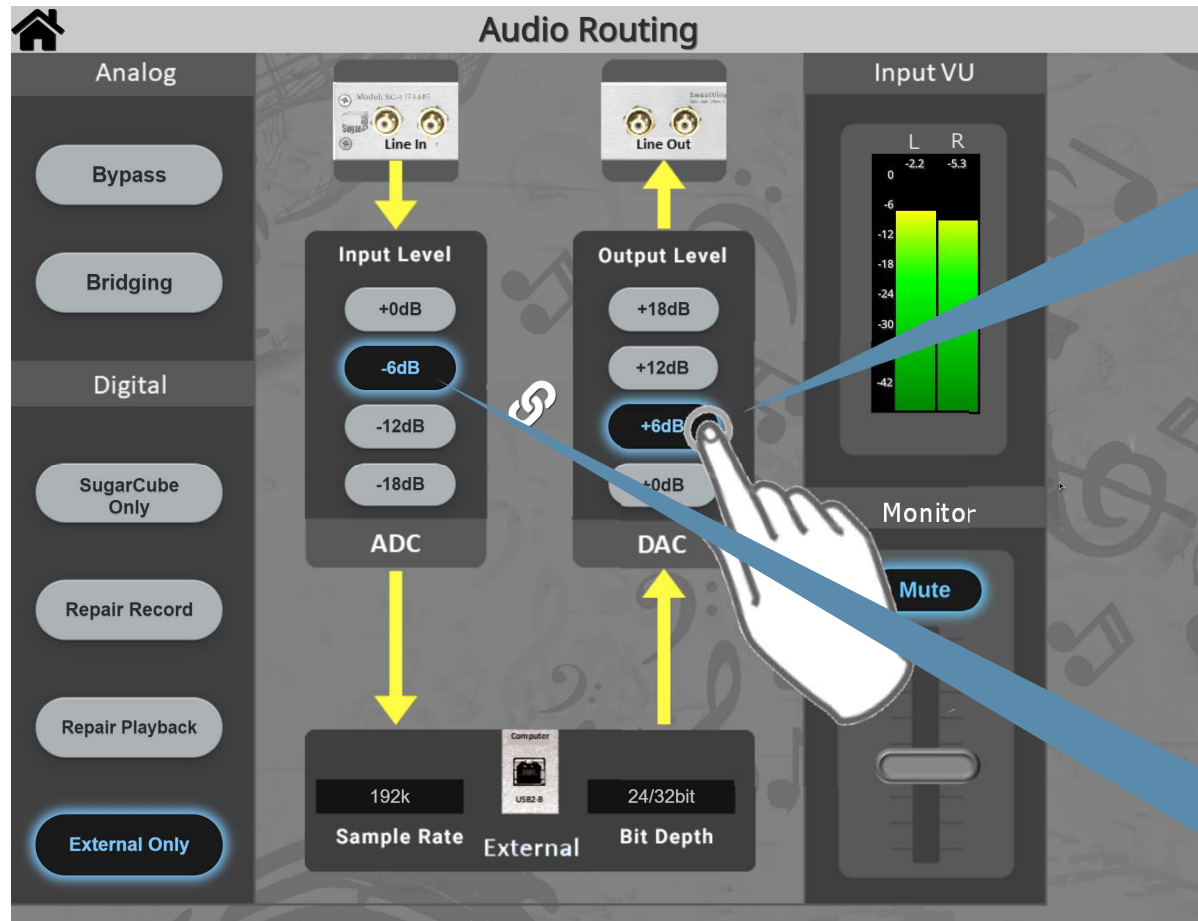
External-Only mode allows an external device to use the **ADC** and **DAC** at **any sample rate**.

Bypass **Bridging** **Mode 1** **Mode 2** **Mode 3** **Mode 4**

In **External Only** mode, the **SC1-Plus** acts like **any other** studio-quality **ADC/DAC** pair and will function at any standard sample rate and bit depth from 192k down to 44.1k and at 16/24/32 bit sample sizes.

Note: Please **press again** the button of **the current mode** of operation if the external device **loses lock** with the SugarCube (you will certainly know due to obvious digital noise.) This resets the internal timing and restores the audio.

5.16 Audio Routing Menu — External Only (Mode 4)



Output Level can be changed during playback to any of the four available levels.

However, input level will change in the **opposite** direction.

Bypass

Bridging

Mode 1

Mode 2

Mode 3

Mode 4

5.17 Audio Routing — External Device **Not** Set to 192k/24bit

The screenshot displays the 'Audio Routing' interface. On the left, a sidebar contains buttons for 'Analog' (Bypass, Bridging), 'Digital' (SugarCube Only, Repair Record, Repair Playback), and 'External Only'. The main area shows 'Line In' and 'Line Out' connections, 'Input Level' (ADC) and 'Output Level' (DAC) settings, and a 'Monitor' section with a VU meter and a 'Mute' button. A yellow box with text points to the 'External' section at the bottom, which shows '96k' for Sample Rate and '24/32bit' for Bit Depth. A red circle highlights the 'SugarCube Only' button in the sidebar. Three callout boxes provide additional information: the top one states that SugarCube repair functions only work with the external device set to 192k/24bit; the middle one explains that if a computer attempts to record or play at a different rate, the system will lock out SugarCube repair functions; the bottom one states that SugarCube repair modes are locked out below 192k/24bit. At the bottom of the interface, a row of mode buttons is shown: Bypass, Bridging, Mode 1, Mode 2, Mode 3, and Mode 4 (which is currently selected).

The **SugarCube** repair functions **only work** with the external device set to **192k/24bit**.

If the connected computer attempts to record or play a digital recording at other than **192k/24bit**, the system will detect this and **lock out SugarCube** repair functions

The SugarCube repair modes are **locked out** below 192k/24bit

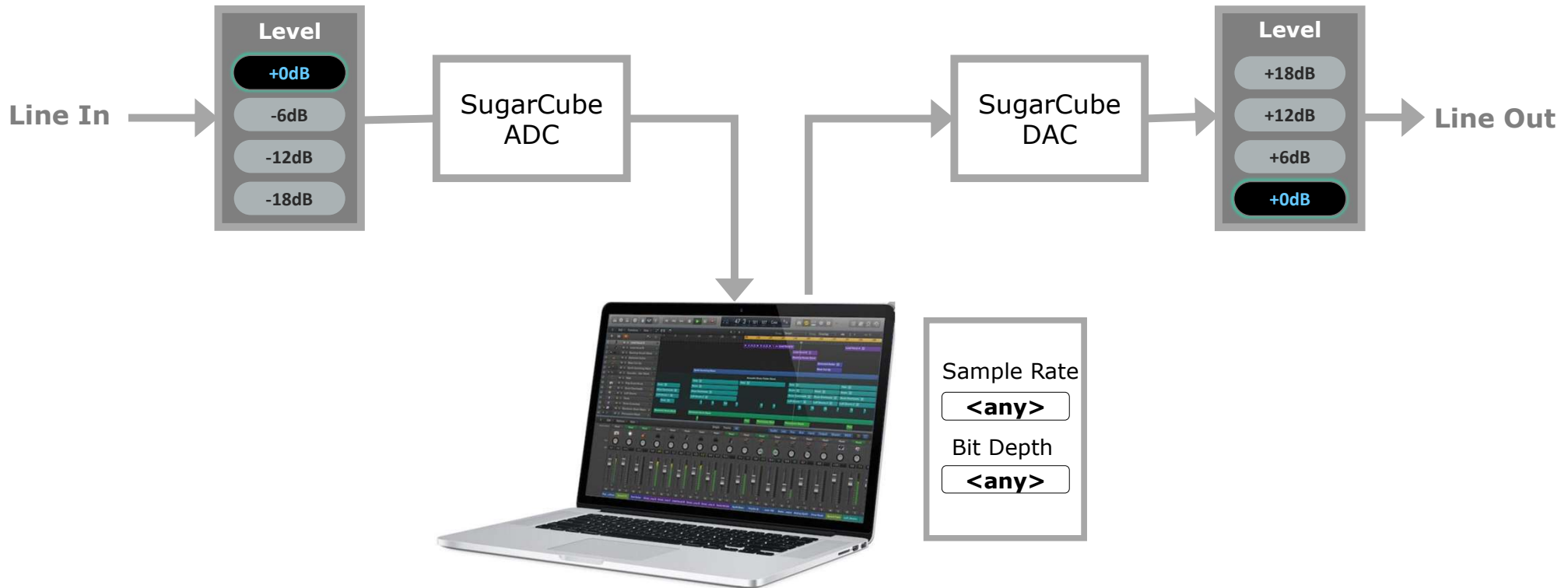
To use SugarCube with an external device, set device to 192k & 24bit

Bypass Bridging Mode 1 Mode 2 Mode 3 **Mode 4**

In **External Only** mode, it is fine to operate at **any sample rate and bit depth below 192k/24bit**, you just cannot simultaneously use the SugarCube repair functions when operating **below 192k/24bit**. 85

5.18 Logical Flow — External-Only (Mode 4)

**SugarCube is out of the loop.
Workstation uses ADC & DAC directly**



Bypass

Bridging

Mode 1

Mode 2

Mode 3

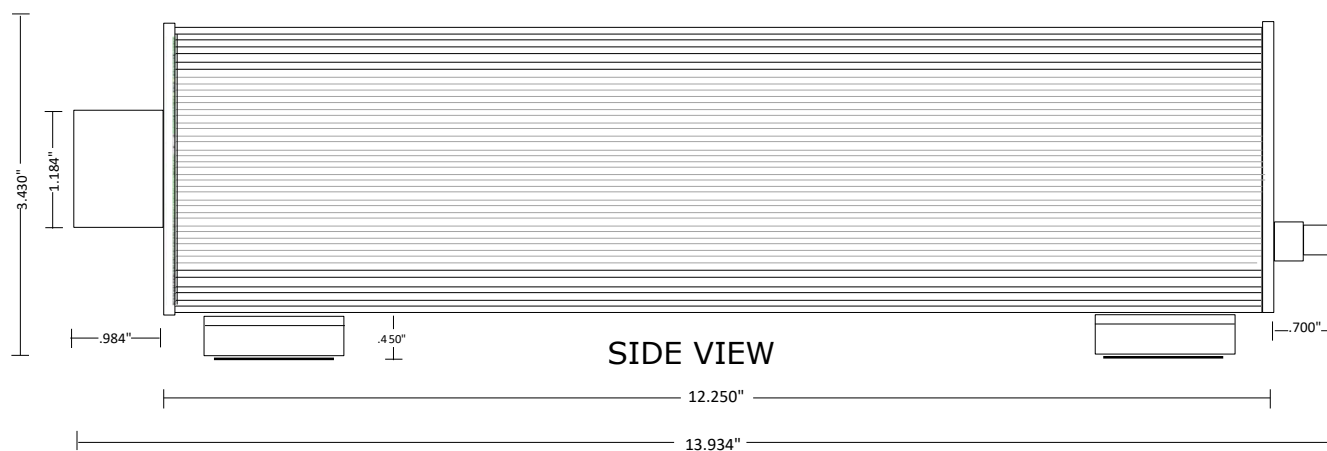
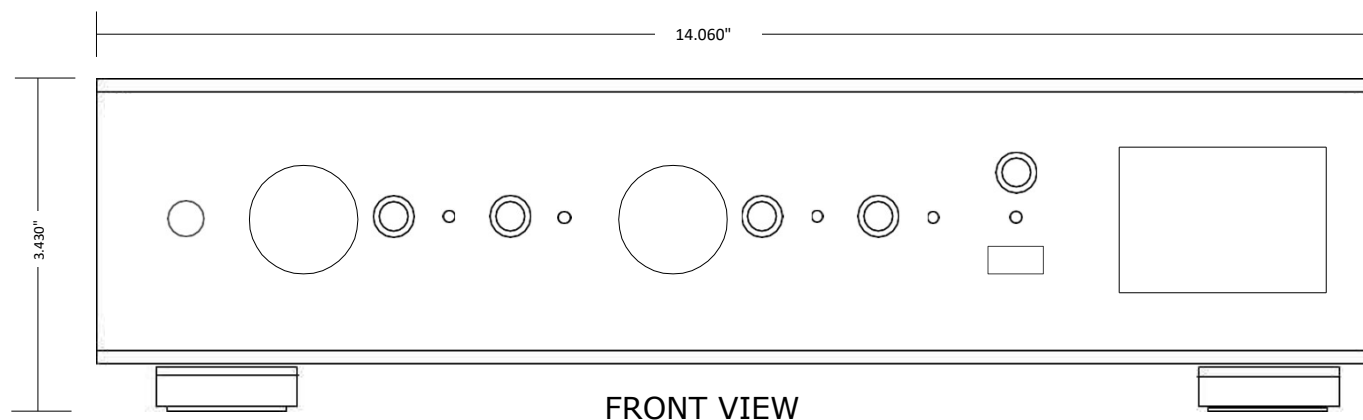
Mode 4

In **External-Only** mode, the SugarCube will operate at **any sample rate and bit depth** from 192k/24bit down to 44.1/16bit, just like any other studio-quality ADC/DAC pair.

Technical Specs



SugarCube SC2-Plus Mechanical:



Technical Specifications

- Line In maximum level = +22.7 dBu
(approx. 10.6V rms or 30.0V p-p)
- Line Out level = Line In level
- THD+N = 0.0031% (A-weighted)
- Frequency Response = 25Hz to 60kHz +/-0.1dB
- Frequency Response = 15Hz to 75kHz -3dB
- DC in = 5 volts, center-pin positive
- Average Power Dissipation = 11 watts

General Info

Support

Thank you for purchasing this SugarCube product.

Please contact customer support at: info@sweetvinyl.com

Trademarks

SweetVinyl™ and SugarCube™ are trademarks of SweetVinyl Inc.
Any non-SweetVinyl trademarks are used for reference purposes only.

Compliance

For the current EU Declaration of conformity, visit:
<http://support.sweetvinyl.com/regulatory>

For regulatory compliance information in North America, visit:
<http://support.sweetvinyl.com/regulatory>

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FAQ

Some common questions
& answers about using
the SC-Plus series



Basic connectivity

- Q: Line-In level setting, why so many choices?
- Q: What is bridging?
- Q: How do I connect through tape loops?
- Q: How do I connecting an external computer?
- Q: My computer only goes to 48k, how to I fix that?

Repair modes

- Q: when I use the noise reduction, do I have to start at the lead in groove or can I play any track on the record?
 - A: This is a yes and no answer: for the first time you play a record using noise reduction (NR), you must start the noise reduction process then drop the needle on the lead in groove.
 - A: Then, if you do not stop the process (that is leave it running), the “fingerprint of the noise that the SugarCube created stays in memory and, as long as you don’t turn off the SugarCube, you can drop the needle anywhere or play other LPs (or 78s) without having to drop the needle in the lead in groove. The NR process will continue to work perfectly.
 - A: Furthermore, if you have LPs from the same label and era (example: late-1950’s Blue Note), the noise “fingerprint” that the SugarCube picked up on the first side of the first LP will work for all other LPs of your Blue Note collection when you use NR.
 - A: However, if you play an LP from say, EMI, that label used a different vinyl plastic formula in making their LPs so their (EMI’s) noise “fingerprint” will be quite different and will require stopping the NR process and re-starting then drop the needle once again in the lead in groove.
 - A: Finally, you could use NR and only one time drop the needle for a “fingerprint” then never do it again (unless you turn the power off) and NR will still work but it will be far from optimal.
 - A: Bottom line: NR is a very powerful tool that can do an amazing job of lowering noise in recordings. It is well worth your time to start and stop NR when the LP is different.

Repair modes

- Q: What does EQ do?
 - LPs are made with a process that boosts the highs and diminishes the lows during the cutting of the master disk. This process was fundamental to record making all the way back to Thomas Edison's days in the early 1900's. However, until the mid to late 1950s, each manufacturer (record label) had their own process that used differing amounts of boost and cut. So, an RCA Victor record would not sound the same played on a non-RCA turntable.
 - More importantly, if you are playing 78s (using a 78-rpm phono cartridge) through your system, your phono pre-amp is almost surely using the RIAA curves which became fully adopted in the early 1960s. RIAA has a much more drastic cut of high-frequencies and a correspondingly drastic boost of the bass frequencies on playback than did any of the 78 recordings made prior to the emergence of the LP in the early 1950s.

Audio routing and use with external devices

- Q: Why can't my computer run Vinyl Studio at any sample rate I want?
- Q: Why would you want to remove clicks & pops during playback instead of during recording.
-